



Completed by:

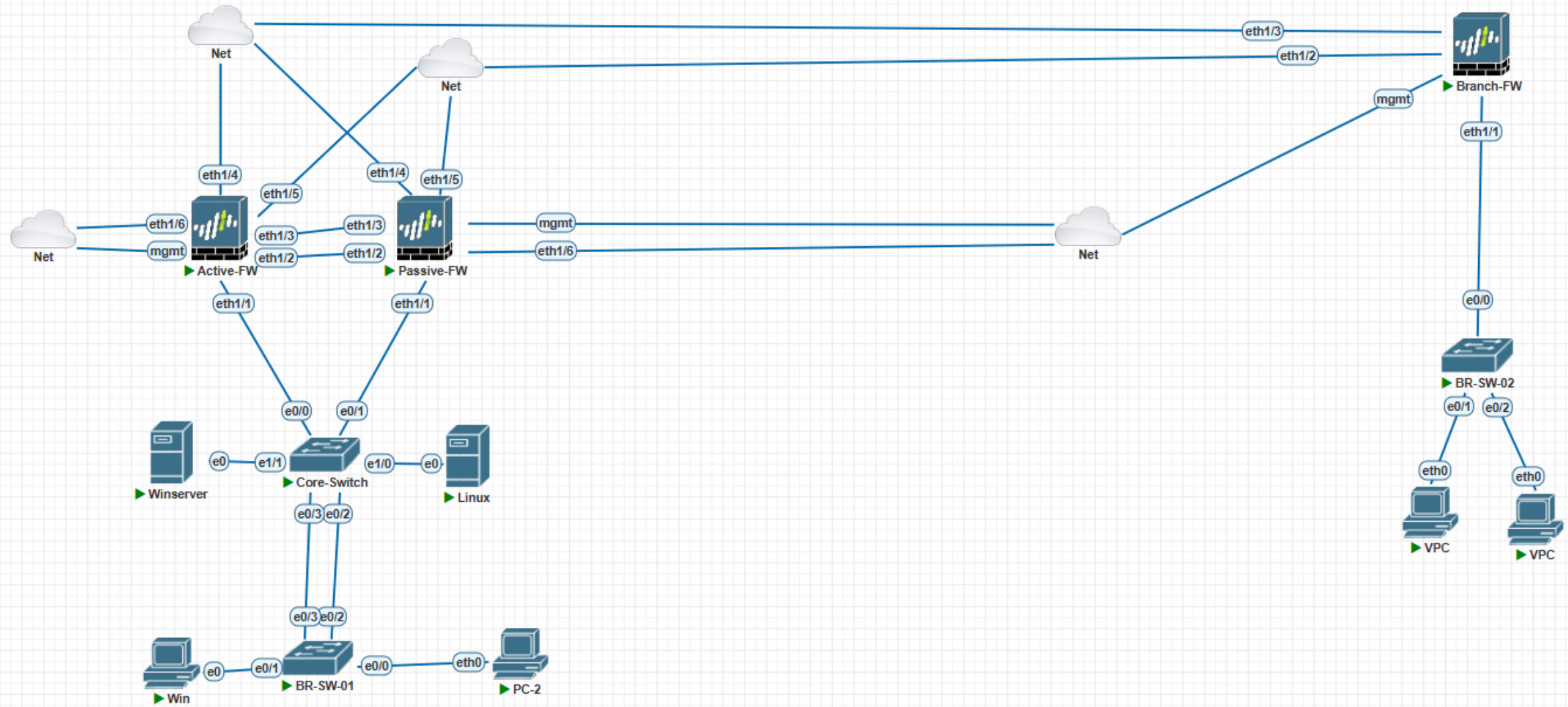
Inal Hajizada

Agaev Kamran

Zeynalli Emil

Development of a Secure Corporate IT Infrastructure with Centralized Monitoring and Management

Delta 172.16.255.0/24
Azintelecom 172.16.254.0/24





Firewall Active

Configuring firewall interfaces

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Portals

Gateways

MDM

Clientless Apps

Clientless App Groups

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

BFD Profile

EthernetVLANLoopbackTunnelSD-WAN

11 items

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE	SD-WAN INTERFACE PROFILE	FEATURES	COMMENT
ethernet1/1	Layer3			ethernet	default	Untagged	none	none			
ethernet1/1.10	Layer3	Allow-Ping		10.1.10.254/24	default	10	none	IT			
ethernet1/1.99	Layer3	Allow-Ping		10.1.99.254/24	default	99	none	Management			
ethernet1/1.100	Layer3	Allow-Ping		10.1.100.254/24	default	100	none	Linux-Servers			
ethernet1/2	HA			none	none	Untagged	none	none			
ethernet1/3	HA			none	none	Untagged	none	none			
ethernet1/4	Layer3	Allow-Ping		172.16.255.1/24	default	Untagged	none	Delta-Private			
ethernet1/5	Layer3	Allow-Ping		172.16.254.1/24	default	Untagged	none	Azintelecom-Private			
ethernet1/6	Layer3	Allow-Ping		172.20.31.201/22	default	Untagged	none	External-Internet			
ethernet1/7				none	none	Untagged	none	none			
ethernet1/8				none	none	Untagged	none	none			

Palo Alto PA-VM dashboard with configured Active-Passive fault tolerance and display of the current device status.

PA-VM

DASHBOARD

ACC

MONITOR

POLICIES

OBJECTS

NETWORK

DEVICE

Layout

3 Columns

Widgets

Last updated

20:20:20

5 mins

High Availability

Mode

Active-passive

Local

Active

Peer 172.20.30.228

Passive

Running Config

Synchronized

App Version

Match

Threat Version

Match

Antivirus Version

Match

PAN-OS Version

Match

GlobalProtect Version

Match

HA1

Up

HA2

Up

Plugin vm_series

Match

General Information

Device Name

PA-VM

MGT IP Address

172.20.28.54 (DHCP)

MGT Netmask

255.255.252.0

MGT Default Gateway

172.20.31.254

MGT IPv6 Address

unknown

MGT IPv6 Link Local Address

fe80::520c:51ff:fe00:1200/64

MGT IPv6 Default Gateway

MGT MAC Address

50:0c:51:00:12:00

Model

PA-VM

Serial #

unknown

CPU ID

KVM:63060000FDFB8B07

Logged In Admins

Admin	From	Client	Session Start	Idle For
admin	172.20.31.77	Web	02/23 06:50:50	00:00:00s
admin	Console	CLI	02/23 08:07:28	00:12:51s

Data Logs

No data available.

System Logs

Description	Time
Connection to Update server: updates.paloaltonetworks.com completed successfully, initiated by 172.20.28.54	02/23 08:18:02
MLAV: Unknown error.	02/23 08:09:37
LOGIN ON tty1 BY admin	02/23 08:07:45
User admin logged in via CLI from Console	02/23 08:07:28
authenticated for user 'admin'. From: Console or telnet.	02/23 08:07:27
Session for user admin via Web from 172.20.29.88 timed out	02/23 08:06:33
Auto update agent found no new IoT updates	02/23 08:06:26
Retrieving Content 'IoT' info failed with error 'The device is not found or not registered, please try after some time '	02/23 08:06:26
Connection to Update server: updates.paloaltonetworks.com completed successfully, initiated by 172.20.28.54	02/23 08:06:25
HA-Sync job succeeded. Completion time=2026/02/23 08:04:08. JobId=4.	02/23 08:04:08

Config Logs

Command	Path	Admin	Time
delete	vsys vsys1 zone Users_PC	admin	02/23 07:38:00
delete	network interface ethernet ethernet1/1 layer3 units ethernet1/1.20	admin	02/23 07:37:25
delete	vsys vsys1 zone Users_PC network layer3 member ethernet1/1.20	admin	02/23 07:37:24
delete	vsys vsys1 import network interface member ethernet1/1.20	admin	02/23 07:37:24
delete	network virtual-router default interface member ethernet1/1.20	admin	02/23 07:37:24
edit	network interface ethernet ethernet1/1 layer3 units ethernet1/1.20	admin	02/23 07:37:12
delete	vsys vsys1 tag 20	admin	02/23 07:36:23
delete	vsys vsys1 address ethernet	admin	02/23 07:35:35
edit	vsys vsys1 address ethernet	admin	02/23 07:35:29
delete	vsys vsys1 address ethernet	admin	02/23 07:34:13

Locks

No locks found

ACC Risk Factor (Last 60 minutes)

0.0

admin

Logout

Last Login Time: 02/23/2026 06:49:29

Session Expire Time: 03/25/2026 07:50:50

Active

Tasks

Language

paloalto

The figure shows virtual tunnel interfaces (tunnels) on the Palo Alto firewall through which VPN traffic flows.

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

InterfacesZonesVLANsVirtual WiresVirtual RoutersIPSec TunnelsGRE TunnelsDHCPDNS ProxyGlobalProtectPortalsGatewaysMDMClientless AppsClientless App GroupsQoSLLDPNetwork ProfilesGlobalProtect IPSec CryptoIKE GatewaysIPSec CryptoIKE CryptoMonitorInterface MgmtZone ProtectionQoS ProfileLLDP ProfileBFD ProfileSD-WAN Interface Profile

EthernetVLANLoopbackTunnelSD-WAN

Q

3 items

INTERFACE	MANAGEMENT PROFILE	IP ADDRESS	VIRTUAL ROUTER	SECURITY ZONE	FEATURES	COMMENT
tunnel		none	none	none		
tunnel.1	Allow-Ping	10.247.139.1/30	default	VPN		
tunnel.2	Allow-Ping	10.247.139.5/30	default	VPN		

The figure shows configured IPsec VPN tunnels on a Palo Alto firewall, providing secure communication between the central office and a remote branch office. Connection parameters are displayed, including IP addresses, interfaces used, and tunnel status.

PA-VM

DASHBOARD
ACC
MONITOR
POLICIES
OBJECTS
NETWORK
DEVICE

Commit

2 items

→

?

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Portals

Gateways

MDM

Clientless Apps

Clientless App Groups

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

BFD Profile

SD-WAN Interface Profile

NAME

STATUS

TYPE

INTERFACE

LOCAL IP

PEER ADDRESS

STATUS

INTERFACE

VIRTUAL ROUTER

VIRTUAL SYSTEM

SECURITY ZONE

STATUS

COMMENT

Delta

● Tunnel Info

Auto Key

ethernet1/4

172.16.255.1/24

172.16.255.254

● IKE Info

tunnel.1

default (Show Routes)

vsys1

VPN

■

Azintelecom

● Tunnel Info

Auto Key

ethernet1/5

172.16.254.1/24

172.16.254.254

● IKE Info

tunnel.2

default (Show Routes)

vsys1

VPN

■

+

 Add

-

 Delete

●

 Enable

○

 Disable

PDF/CSV

admin

Logout

Last Login Time: 02/27/2026 04:17:40

Session Expire Time: 03/29/2026 05:19:17

Active

Tasks

Language

paloalto

IKE gateway settings for VPN connections.

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

2 items

			Local Address		Peer ID		Local ID			IKE Advanced Options				
	NAME	PEER ADDRESS	INTERFACE	IP	ID	TYPE	ID	TYPE	VERSION	MODE	PASSIVE MODE	NAT TRAVERSAL	CRYPTO PROFILE	
☐	DELTA	172.16.255.254	ethernet1/4	172.16.255.1/24					ikev2	auto	☐	☐	Delta-IKE	
☐	Azintelecom	172.16.254.254	ethernet1/5	172.16.254.1/24					ikev2	auto	☐	☐	Azintelecom-IKE	

+ Add

- Delete

Enable

Disable

PDF/CSV

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Portals

Gateways

MDM

Clientless Apps

Clientless App Groups

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

BFD Profile

SD-WAN Interface Profile

The figure shows the IPSec Crypto settings on the Palo Alto firewall for secure VPN connections, i.e., how the packet will be packaged in which type of encryption.

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Portals

Gateways

MDM

Clientless Apps

Clientless App Groups

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

BFD Profile

SD-WAN Interface Profile

5 items

NAME	ESP/AH	ENCRYPTION	AUTHENTICATION	DH GROUP	LIFETIME	LIFESIZE
default	ESP	aes-128-cbc, 3des	sha1	group2	1 hours	
Suite-B-GCM-128	ESP	aes-128-gcm	none	group19	1 hours	
Suite-B-GCM-256	ESP	aes-256-gcm	none	group20	1 hours	
Delta-IPSEC	ESP	des	md5	group1	1 hours	
Azintelecom-IPSEC	ESP	des	md5	group1	1 hours	

+

 Add

-

 Delete

↻

 Clone

PDF/CSV

DHCP server settings on the Palo Alto firewall, providing automatic assignment of IP addresses, gateway, and DNS for VLAN 10

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Portals

Gateways

MDM

Clientless Apps

Clientless App Groups

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

BFD Profile

SD-WAN Interface Profile

DHCP Server

DHCP Relay

1 item

INTERFACE ^	MODE	PROBE IP	OPTIONS	IP POOLS	RESERVED
☐ ethernet1/1.10	auto	☐	Lease: Unlimited DNS: 8.8.8.8 Gateway: 10.1.10.254	View Allocation 10.1.10.10-10.1.10.200	

Add

Delete

PDF/CSV

admin | Logout | Last Login Time: 02/23/2026 06:49:29 | Session Expire Time: 03/25/2026 07:50:50

Active | Tasks | Language | paloalto

We configured HA for fault tolerance, Active and Passive. HA1 carries service information, Hello packets, etc. HA2 is needed for synchronization.

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

SetupHigh AvailabilityConfig AuditPassword ProfilesAdministratorsAdmin RolesAuthentication ProfileAuthentication SequenceUser IdentificationData RedistributionDevice QuarantineVM Information SourcesTroubleshootingCertificate ManagementCertificatesCertificate ProfileOCSP ResponderSSL/TLS Service ProfileSCEPSSL Decryption ExclusivSSH Service ProfileResponse PagesLog SettingsServer ProfilesSNMP TrapSyslogEmailHTTPNetflowRADIUS

GeneralHA CommunicationsLink and Path MonitoringOperational Commands

Control Links

HA1

Port ethernet1/2IPv4/IPv6 Address 10.10.10.1Netmask 255.255.255.252GatewayEncryption Enabled ☐

HA1 Backup

PortIPv4/IPv6 AddressNetmaskGateway

Data Links

HA2

Enable Session Synchronization ☒Port ethernet1/3IPv4/IPv6 AddressNetmaskGatewayTransport ethernetAction log-onlyThreshold (ms) 10000

HA2 Backup

PortIPv4/IPv6 AddressNetmaskGateway

Packet Forwarding

HA3 InterfaceVR Sync ☐QoS Sync ☐Tentative Hold Time (sec) 60

Remove All

We configured link and monitoring so that HA would switch over not only when the firewall died, but also when the cable broke.

Setup

High Availability

Config Audit

Password Profiles

Administrators

Admin Roles

Authentication Profile

Authentication Sequence

User Identification

Data Redistribution

Device Quarantine

VM Information Sources

Troubleshooting

Certificate Management

Certificates

Certificate Profile

OCSP Responder

SSL/TLS Service Profile

SCEP

SSL Decryption Exclusion

SSH Service Profile

Response Pages

Log Settings

Server Profiles

SNMP Trap

Syslog

Email

HTTP

Netflow

RADIUS

General

HA Communications

Link and Path Monitoring

Operational Commands

Link Monitoring

Enabled ☒

Failure Condition any

Link Group

☐	NAME	ENABLED	GROUP FAILURE CONDITION	INTERFACES
☒	Main	☒	any	ethernet1/1

+ Add

- Delete

Path Monitoring

Enabled ☒

Failure Condition any

Path Group

☐	NAME	TYPE	ENAB...	FAILURE CONDITION	SOURCE IP	DESTINATION IP GROUP	PING INTERVAL	PING COUNT
--------------------------	------	------	---------	-------------------	-----------	----------------------	---------------	------------

+ Add Virtual Wire Path

+ Add VLAN Path

+ Add Virtual Router Path

- Delete

- Remove All

admin

Logout

Last Login Time: 02/23/2026 06:49:29

Session Expire Time: 03/25/2026 07:50:50

Active

Tasks

Language

naloalto

The figure shows the syslog configuration with the Palo Alto firewall on the Wazuh monitoring server for collecting and analyzing security events.

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

Device Quarantine

VM Information Sources

Troubleshooting

Certificate Management

- Certificates
- Certificate Profile
- OCSP Responder
- SSL/TLS Service Profile
- SCEP
- SSL Decryption Exclusion
- SSH Service Profile

Response Pages

Log Settings

Server Profiles

- SNMP Trap
- Syslog
- Email
- HTTP
- Netflow
- RADIUS
- TACACS+
- LDAP
- Kerberos
- SAML Identity Provider
- Multi Factor Authentication

Local User Database

- Users
- User Groups

Scheduled Log Export

Software

1 item

Servers								
NAME	LOCATION	NAME	SYSLOG SERVER	TRANSPORT	PORT	FORMAT	FACILITY	
☐	wazuh		wazuh-server	172.20.28.58	UDP	514	BSD	LOG_USER

Add

Delete

Clone

PDF/CSV



Firewall Passive

Dashboard and status of the Palo Alto HA cluster.

← → ↺ ⚠ Not secure 172.20.30.228/?#dashboard:vsys1

PA-VM

DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK DEVICE

Layout 3 Columns Widgets Last updated 20:04:59 5 mins

High Availability

Mode Active-passive

Local ● Passive

Peer [172.20.28.54](#) ● Active

Running Config ● Synchronized

App Version ● Match

Threat Version ● Match

Antivirus Version ● Match

PAN-OS Version ● Match

GlobalProtect Version ● Match

HA1 ● Up

HA2 ● Up

Plugin vm_series ● Match

General Information

Device Name PA-VM

MGT IP Address 172.20.30.228 (DHCP)

MGT Netmask 255.255.252.0

MGT Default Gateway 172.20.31.254

MGT IPv6 Address unknown

MGT IPv6 Link Local Address fe80::52e5:c4ff:fe00:e00/64

MGT IPv6 Default Gateway

MGT MAC Address 50:e5:c4:00:0e:00

Model PA-VM

Serial # unknown

CPU ID KVM:63060000FDFB8B07

Logged In Admins

Admin	From	Client	Session Start	Idle For
admin	172.20.31.77	Web	02/23 06:50:17	00:00:00s
admin	172.20.29.88	Web	02/23 07:07:02	00:57:43s

Data Logs

No data available.

System Logs

Description	Time
HA Group 10: Threat Content version now matches	02/23 08:04:07
HA Group 10: Threat Content version mismatch due to device update	02/23 08:04:07
HA Group 10: Anti-Virus version now matches	02/23 08:04:07
HA Group 10: Anti-Virus version mismatch due to device update	02/23 08:04:07
HA Group 10: Application Content version now matches	02/23 08:04:07
HA Group 10: Application Content version mismatch due to device update	02/23 08:04:07
SSLMGR daemon not sync to HA peer.for device block list	02/23 08:04:05
SSLMGR daemon not sync to HA peer.for certificate status	02/23 08:04:05
SSLMGR daemon not sync to HA peer.for satellite info	02/23 08:04:05
running configuration synchronized with HA peer by admin	02/23 08:03:44

Config Logs

No data available.

Locks

No locks found

ACC Risk Factor (Last 60 minutes)

0.0

Link and path monitoring activated; if the active firewall's cables break, etc., the passive firewall becomes active.

← → ↺

Not secure 172.20.30.228/?#device::vsys1::device/high-availability

☆ | 📁 | ⬇️ | E | ⋮

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

↺ ?

Setup

High Availability

Config Audit

Password Profiles

Administrators

Admin Roles

Authentication Profile

Authentication Sequence

User Identification

Data Redistribution

Device Quarantine

VM Information Sources

Troubleshooting

Certificate Management

Certificates

Certificate Profile

OCSP Responder

SSL/TLS Service Profile

SCEP

SSL Decryption Exclusion

SSH Service Profile

Response Pages

Log Settings

Server Profiles

SNMP Trap

Syslog

Email

HTTP

Netflow

RADIUS

General

HA Communications

Link and Path Monitoring

Operational Commands

Link Monitoring

Enabled ☒

Failure Condition any

Link Group

☐	NAME	ENABLED	GROUP FAILURE CONDITION	INTERFACES
☐	Main	☒	any	ethernet1/1

+ Add

- Delete

Path Monitoring

Enabled ☒

Failure Condition any

Path Group

☐	NAME	TYPE	ENAB...	FAILURE CONDITION	SOURCE IP	DESTINATION IP GROUP	PING INTERVAL	PING COUNT
--------------------------	------	------	---------	-------------------	-----------	----------------------	---------------	------------

+ Add Virtual Wire Path

+ Add VLAN Path

+ Add Virtual Router Path

- Delete

Remove All



Branch-2 Firewall

Configuring firewall interfaces

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Portals

Gateways

MDM

Clientless Apps

Clientless App Groups

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

BFD Profile

SD-WAN Interface Profile

EthernetVLANLoopbackTunnelSD-WAN

9 items

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE	SD-WAN INTERFACE PROFILE	FEATURES	COMMENT
ethernet1/1	Layer3			none	none	Untagged	none	none			
ethernet1/1.10	Layer3	Allow-Ping		10.2.10.254/24	default	10	none	IT			
ethernet1/2	Layer3	Allow-Ping		172.16.254.254/24	default	Untagged	none	Azintelecom-Private			
ethernet1/3	Layer3	Allow-Ping		172.16.255.254/24	default	Untagged	none	Delta-Private			
ethernet1/4				none	none	Untagged	none	none			
ethernet1/5				none	none	Untagged	none	none			
ethernet1/6				none	none	Untagged	none	none			
ethernet1/7				none	none	Untagged	none	none			
ethernet1/8				none	none	Untagged	none	none			

+

 Add Subinterface

-

 Delete

PDF/CSV

Settings for zones that separate the internal network, VPN, and provider channels

PA-VM

DASHBOARD
ACC
MONITOR
POLICIES
OBJECTS
NETWORK
DEVICE

Commit
|
|
|

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Portals

Gateways

MDM

Clientless Apps

Clientless App Groups

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

BFD Profile

SD-WAN Interface Profile

Q

4 items

→

×

							User-ID			Device-ID		
	NAME	TYPE	INTERFACES / VIRTUAL SYSTEMS	ZONE PROTECTION PROFILE	PACKET BUFFER PROTECTION	LOG SETTING	ENABLED	INCLUDED NETWORKS	EXCLUDED NETWORKS	ENABLED	INCLUDED NETWORKS	EXCLUDED NETWORKS
☐	Azintelecom-Private	layer3	ethernet1/2		☒		☐	any	none	☐	any	none
☐	Delta-Private	layer3	ethernet1/3		☒		☐	any	none	☐	any	none
☐	IT	layer3	ethernet1/1.10		☒		☐	any	none	☐	any	none
☐	VPN	layer3	tunnel.1		☒		☐	any	none	☐	any	none
			tunnel.2									

+

 Add

-

 Delete

PDF/CSV

admin

Logout

Last Login Time: 02/23/2026 06:51:01

Session Expire Time: 03/25/2026 09:12:36

Tasks
Language
paloalto

Tunnel settings, including interfaces, IP addresses, connection status.

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

2 items

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Portals

Gateways

MDM

Clientless Apps

Clientless App Groups

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

bfd Profile

SD-WAN Interface Profile

NAME

STATUS

TYPE

IKE Gateway/Satellite

Tunnel Interface

COMMENT

Delta

Tunnel Info

Auto Key

ethernet1/3

172.16.255.254/...

172.16.255.1

IKE Info

tunnel.1

default (Show Routes)

vsys1

VPN

Azintelecom

Tunnel Info

Auto Key

ethernet1/2

172.16.254.254/...

172.16.254.1

IKE Info

tunnel.2

default (Show Routes)

vsys1

VPN

+

Add

-

Delete

✓

Enable

✗

Disable

PDF/CSV

admin

Logout

Last Login Time: 02/23/2026 06:51:01

Session Expire Time: 03/25/2026 09:12:36

Tasks

Language

paloalto

DHCP server settings on the Palo Alto firewall, providing automatic assignment of IP addresses, gateway, and DNS for VLAN 10

PA-VM

DASHBOARDACCMONITORPOLICIESOBJECTSNETWORKDEVICE

Commit

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Portals

Gateways

MDM

Clientless Apps

Clientless App Groups

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

BFD Profile

SD-WAN Interface Profile

DHCP Server

DHCP Relay

1 item

INTERFACE	MODE	PROBE IP	OPTIONS	IP POOLS	RESERVED
☐ ethernet1/1.10	auto	☐	Lease: Unlimited DNS: 8.8.8.8 Gateway: 10.2.10.254	View Allocation 10.2.10.10-10.2.10.200	

Add

Delete

PDF/CSV



Core Switch

Here, Etherchannel is configured on ports E0/2, E0/3, trunks, accesses, DTP, DHCP Snooping, STP, and Port Security are disabled.

```
!
interface Port-channel1
switchport trunk allowed vlan 10,20,30
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
!
interface Ethernet0/0
description Link-To-Palo-Alto
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Ethernet0/1
description Link-To-Palo-Alto
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Ethernet0/2
switchport trunk allowed vlan 10,20,30
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
channel-group 1 mode active
!
interface Ethernet0/3
switchport trunk allowed vlan 10,20,30
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
channel-group 1 mode active
!
interface Ethernet1/0
switchport access vlan 100
switchport mode access
switchport nonegotiate
switchport port-security mac-address sticky
switchport port-security mac-address sticky 50a7.e400.1800
switchport port-security
spanning-tree portfast edge
spanning-tree bpduguard enable
!
interface Ethernet1/1
switchport access vlan 10
switchport mode access
switchport port-security mac-address sticky
switchport port-security mac-address sticky 5054.5500.1000
switchport port-security
spanning-tree portfast edge
!
```




BR-SW-01

|

Here, too, Etherchannel is configured on ports E0/2, E0/3, trunks, accesses, DTP, DHCP Snooping, STP, and Port Security are disabled.

```
!
interface Port-channel1
switchport trunk allowed vlan 10,20,30
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
!
interface Ethernet0/0
switchport access vlan 10
switchport mode access
switchport nonegotiate
switchport port-security mac-address sticky
switchport port-security
!
interface Ethernet0/1
switchport access vlan 20
switchport mode access
switchport nonegotiate
switchport port-security mac-address sticky
switchport port-security mac-address sticky 500b.dd00.1300
switchport port-security
!
interface Ethernet0/2
switchport trunk allowed vlan 10,20,30
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
channel-group 1 mode active
!
interface Ethernet0/3
switchport trunk allowed vlan 10,20,30
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
channel-group 1 mode active
!
interface Ethernet1/0
!
interface Ethernet1/1
!
interface Ethernet1/2
!
interface Ethernet1/3
!
ip forward-protocol nd
!
ip http server
!
```



BR-SW-02

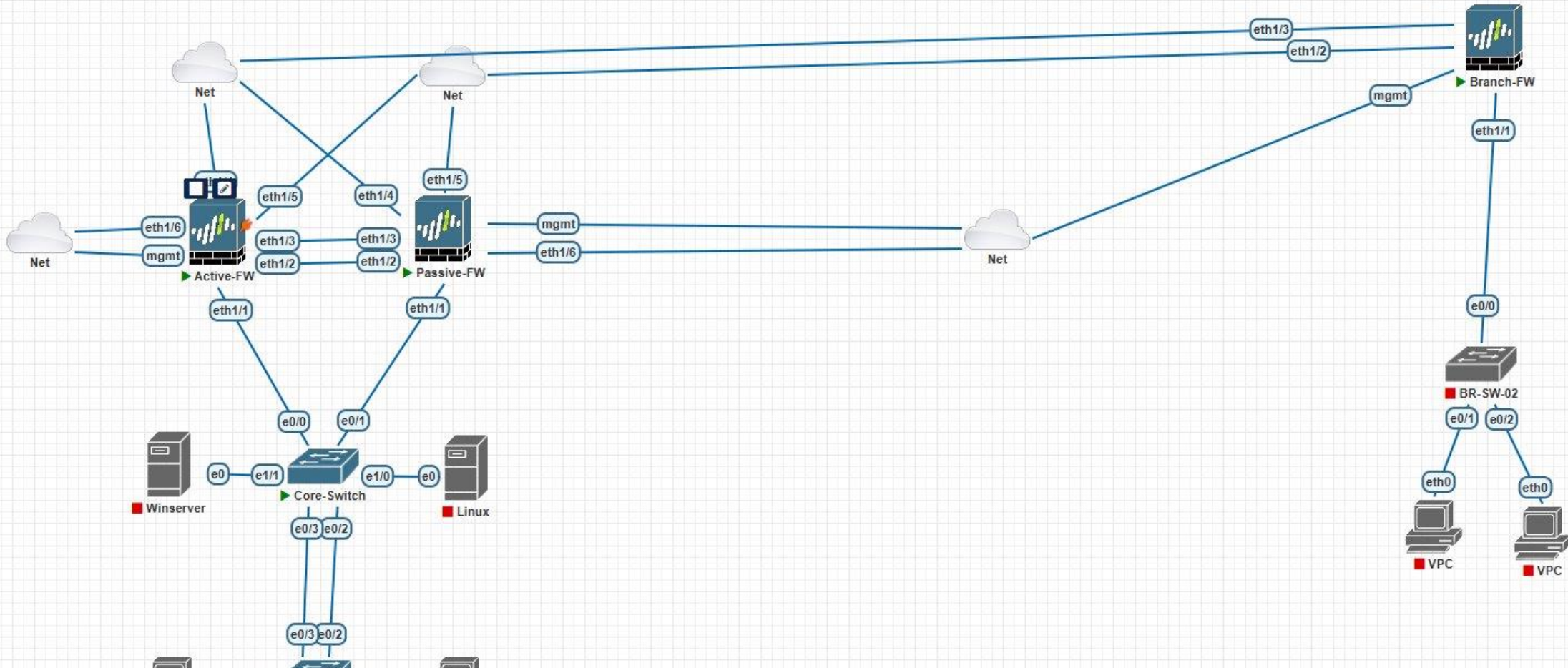
|

```
ip dhcp snooping trust
!
interface Ethernet0/1
 switchport access vlan 10
 switchport mode access
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0050.7966.6803
 switchport port-security
 ip arp inspection limit rate 20
 spanning-tree portfast edge
 spanning-tree bpduguard enable
 ip dhcp snooping limit rate 15
!
interface Ethernet0/2
 switchport access vlan 10
 switchport mode access
 switchport port-security
 ip arp inspection limit rate 10
 spanning-tree portfast edge
 spanning-tree bpduguard enable
 ip dhcp snooping limit rate 15
!
interface Ethernet0/3
!
ip forward-protocol nd
!
ip http server
!
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
!
!
!
!
control-plane
!
!
line con 0
 logging synchronous
line aux 0
line vty 0 4
 login
!
!
!
end
l2-sw(config)#
```

```
interface Ethernet0/0
switchport access vlan 10
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
ip arp inspection trust
ip dhcp snooping trust
!
interface Ethernet0/1
switchport access vlan 10
switchport mode access
switchport port-security
ip arp inspection limit rate 20
spanning-tree portfast edge
spanning-tree bpduguard enable
ip dhcp snooping limit rate 15
!
interface Ethernet0/2
switchport access vlan 10
switchport mode access
switchport port-security
ip arp inspection limit rate 10
spanning-tree portfast edge
spanning-tree bpduguard enable
ip dhcp snooping limit rate 15
!
interface Ethernet0/3
ip forward-protocol nd
ip http server
ip ssh server algorithm encryption aes128-ctr aes192-ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-ctr aes256-ctr
!
control-plane
!
line con 0
logging synchronous
line aux 0
line vty 0 4
login
```

Windows Server / Active Directory

Delta 172.16.255.0/24
Azintelecom 172.16.254.0/24





Server Manager

Add Roles and Features Wizard

Select destination server

DESTINATION SERVER
srv

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool
☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
srv	169.254.140.239	Microsoft Windows Server 2012 R2 Standard

1 Computer(s) found

This page shows servers that are running Windows Server 2012, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

< Previous Next > Install Cancel

Events
Performance
BPA results

1 Events
Services
Performance

5:05 PM

Server Manager

Server Manager ▸ Dashboard

Manage Tools View Help

Dashboard

Local Server

All Servers

AD DS

DNS

File and Storage

WELCOME TO SERVER MANAGER

Add Roles and Features Wizard

Installation progress

DESTINATION SERVER
srv

Before You Begin

Installation Type

Server Selection

Server Roles

Features

AD DS

DNS Server

Confirmation

Results

View installation progress

Feature installation

Configuration required. Installation succeeded on srv.

Active Directory Domain Services

Additional steps are required to make this machine a domain controller.
[Promote this server to a domain controller](#)

DNS Server

Group Policy Management

Remote Server Administration Tools

Role Administration Tools

AD DS and AD LDS Tools

Active Directory module for Windows PowerShell

AD DS Tools

Active Directory Administrative Center

You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.

[Export configuration settings](#)

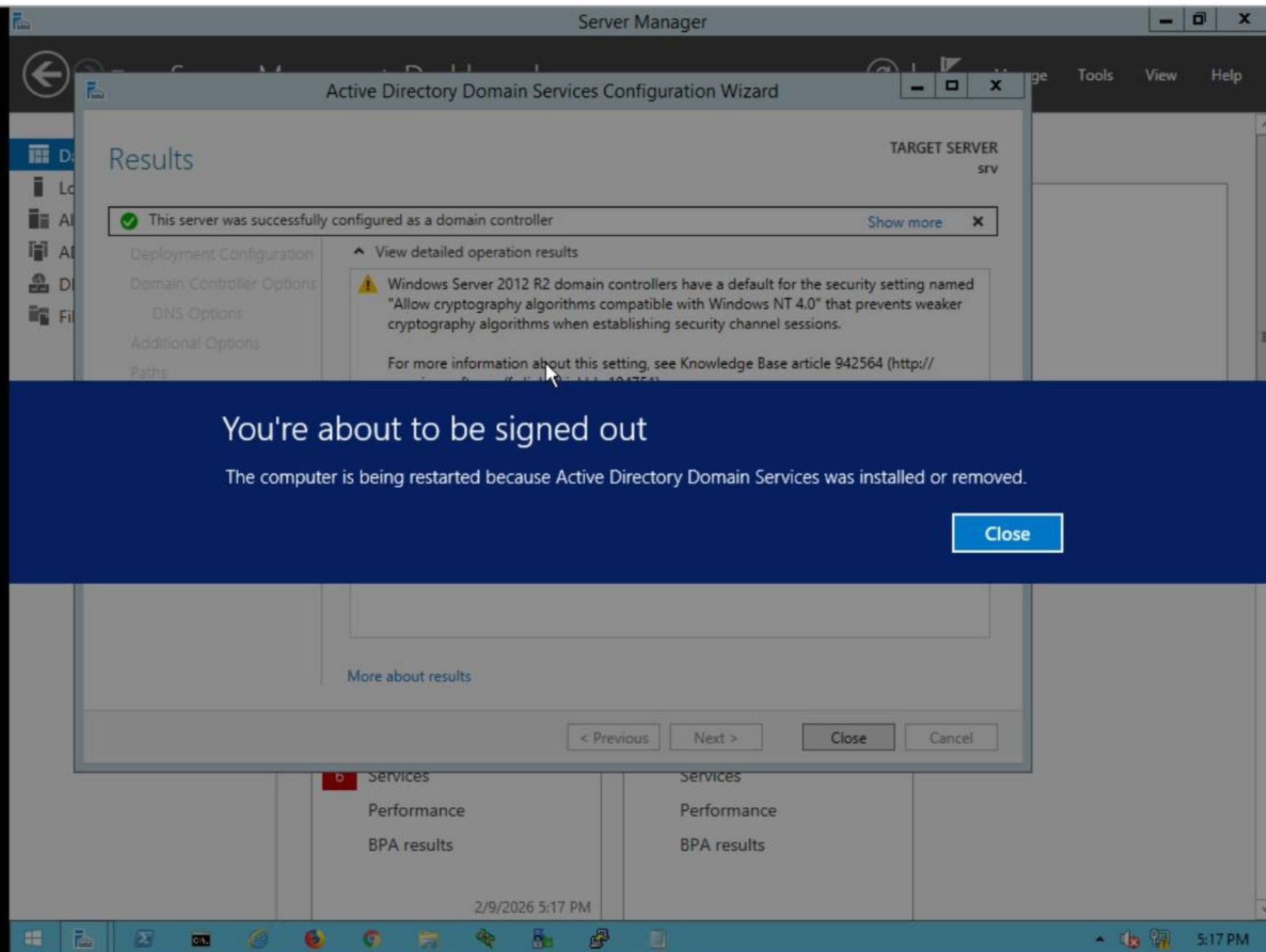
< Previous

Next >

Close

Cancel

Hide



Server Manager

Server Manager Dashboard

Manage Tools View Help

Local Server

All Servers

AD DS

DNS

File and Storage Services

Computer Name/Domain Changes

You can change the name and the membership of this computer. Changes might affect access to network resources.

Computer name:
KEI-DC01

Full computer name:
KEI-DC01.KEI.LOCAL

More...

Member of

☒ Domain:
KEI.LOCAL

☐ Workgroup:

OK Cancel

DNS 1

Manageability

Events

Services

Performance

BPA results

Local Server 1

Manageability

1 Events

Services

Performance

BPA results

2/9/2026 5:21 PM

Events

Services

Performance

BPA results

ADMX

Administrative Templates: Policy definitions (ADMX files) retrieved from the central store.

PolicyDefinitions				
File	Home	Share	View	
Network > KEI-DC01 > SYSVOL > KEI.LOCAL > Policies > PolicyDefinitions				
Search PolicyDefinitions				
Name	Date modified	Type	Size	
en-US	2/18/2026 6:07 PM	File folder		
ActiveXInstallService.admx	6/18/2013 8:07 PM	ADMX File	5 KB	
AddRemovePrograms.admx	6/18/2013 7:57 PM	ADMX File	5 KB	
AdmPwd.admx	4/28/2021 2:56 PM	ADMX File	4 KB	
AppCompat.admx	6/18/2013 7:44 PM	ADMX File	6 KB	
AppxPackageManager.admx	6/18/2013 7:46 PM	ADMX File	3 KB	
AppXRuntime.admx	7/18/2013 8:53 PM	ADMX File	4 KB	
AttachmentManager.admx	6/18/2013 7:45 PM	ADMX File	6 KB	
AuditSettings.admx	6/18/2013 8:07 PM	ADMX File	2 KB	
AutoPlay.admx	6/18/2013 7:57 PM	ADMX File	4 KB	
Biometrics.admx	6/18/2013 7:57 PM	ADMX File	3 KB	
Bits.admx	6/18/2013 8:02 PM	ADMX File	56 KB	
CEIPEnable.admx	6/18/2013 8:12 PM	ADMX File	2 KB	
CipherSuiteOrder.admx	6/18/2013 7:57 PM	ADMX File	2 KB	
COM.admx	6/18/2013 7:55 PM	ADMX File	2 KB	
Conf.admx	6/18/2013 7:48 PM	ADMX File	14 KB	
ControlPanel.admx	6/18/2013 7:57 PM	ADMX File	3 KB	
ControlPanelDisplay.admx	6/18/2013 7:57 PM	ADMX File	15 KB	
Cpls.admx	6/18/2013 7:57 PM	ADMX File	2 KB	
CredentialProviders.admx	6/18/2013 7:57 PM	ADMX File	4 KB	
CredSsp.admx	6/18/2013 8:03 PM	ADMX File	12 KB	
CredUI.admx	6/18/2013 7:57 PM	ADMX File	3 KB	
CtrlAltDel.admx	6/18/2013 8:08 PM	ADMX File	3 KB	
DCOM.admx	6/18/2013 7:55 PM	ADMX File	3 KB	
Desktop.admx	6/18/2013 7:57 PM	ADMX File	14 KB	
DeviceCompat.admx	6/18/2013 7:44 PM	ADMX File	2 KB	
DeviceInstallation.admx	6/18/2013 8:27 PM	ADMX File	13 KB	

LAPS

The figure shows PowerShell commands being executed on a Windows Server to delegate rights to computer objects in Active Directory. The Set-AdmPwdComputerSelfPermission command is used to configure permissions for the organizational units (OU) of Branch A and Branch B, allowing computers in these units to independently update and manage their local administrator passwords within the LAPS solution. The Delegated status confirms that delegation has been successfully applied.

```
PS C:\Users\Administrator> Set-AdmPwdComputerSelfPermission -OrgUnit "OU=Computers, OU=Branch A, DC=KEI, DC=LOCAL"

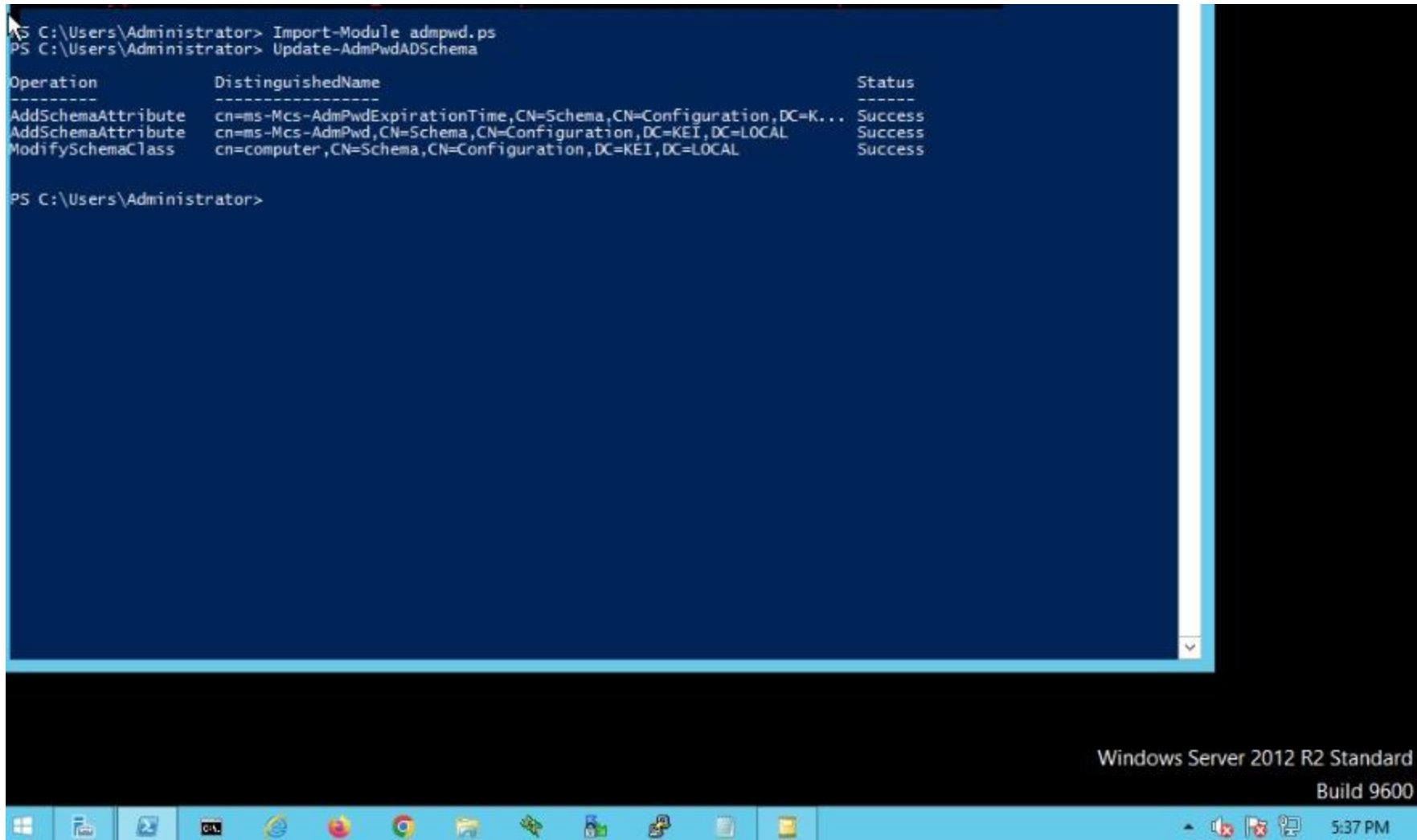
Name           DistinguishedName           Status
----           -
Computers      OU=Computers,OU=Branch A,DC=KEI,DC=LOCAL  Delegated

PS C:\Users\Administrator> Set-AdmPwdComputerSelfPermission -OrgUnit "OU=Computers, OU=Branch B, DC=KEI, DC=LOCAL"

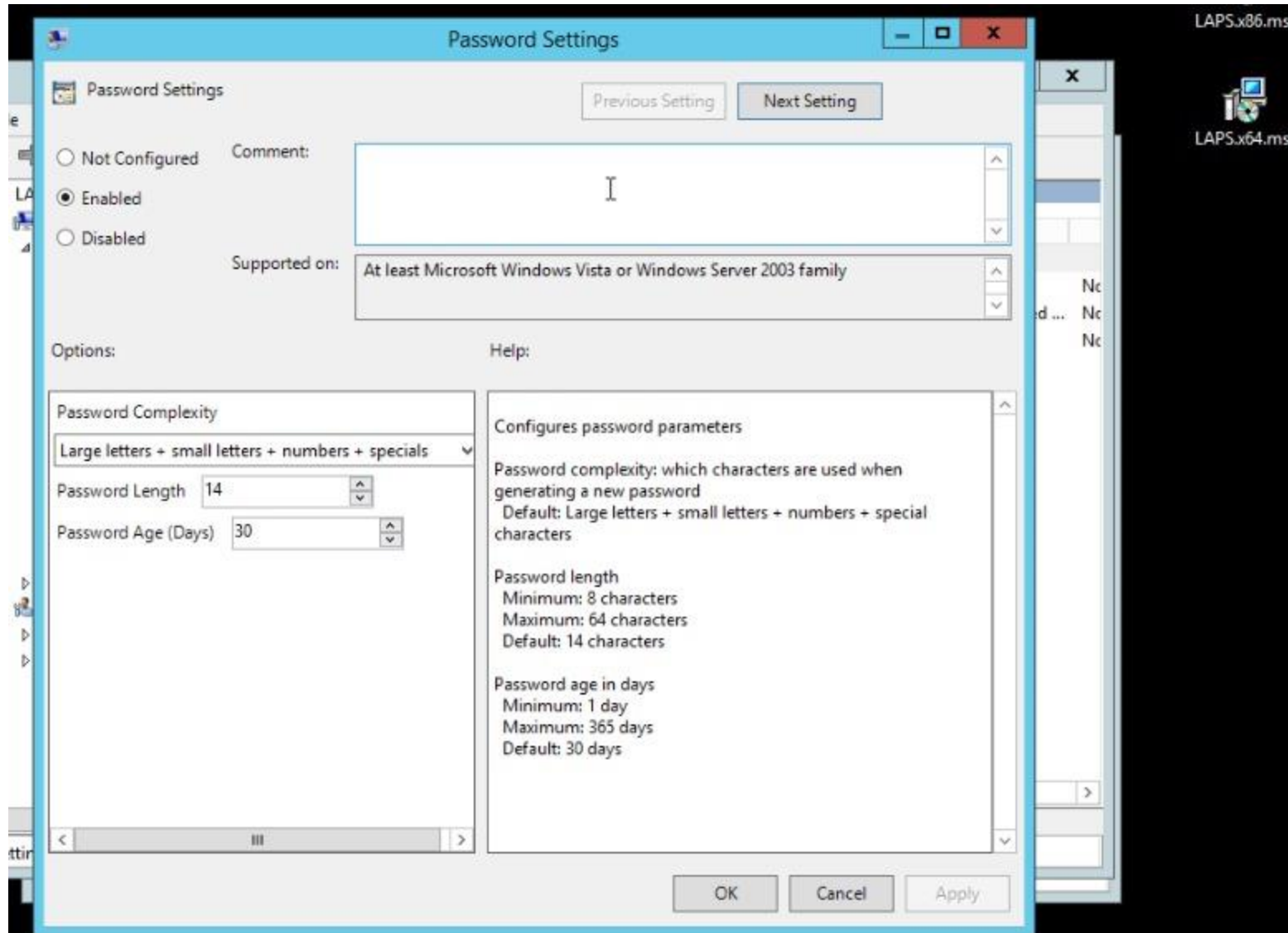
Name           DistinguishedName           Status
----           -
Computers      OU=Computers,OU=Branch B,DC=KEI,DC=LOCAL  Delegated

PS C:\Users\Administrator>
```


The Import-Module admPwd.ps command loads the LAPS management module, after which the Update-AdmPwdADSchema command extends the Active Directory schema. As a result, special attributes (ms-Mcs-AdmPwd and ms-Mcs-AdmPwdExpirationTime) are added to the schema to store the unique local administrator password and its expiration time for each domain computer. The Success status confirms that the schema has been successfully updated.



This policy defines the requirements for automatically generated passwords for local administrator accounts on domain computers. The setting is enabled, which means that the policy is actively applied. Password complexity requirements are set.



 LAPS UI 

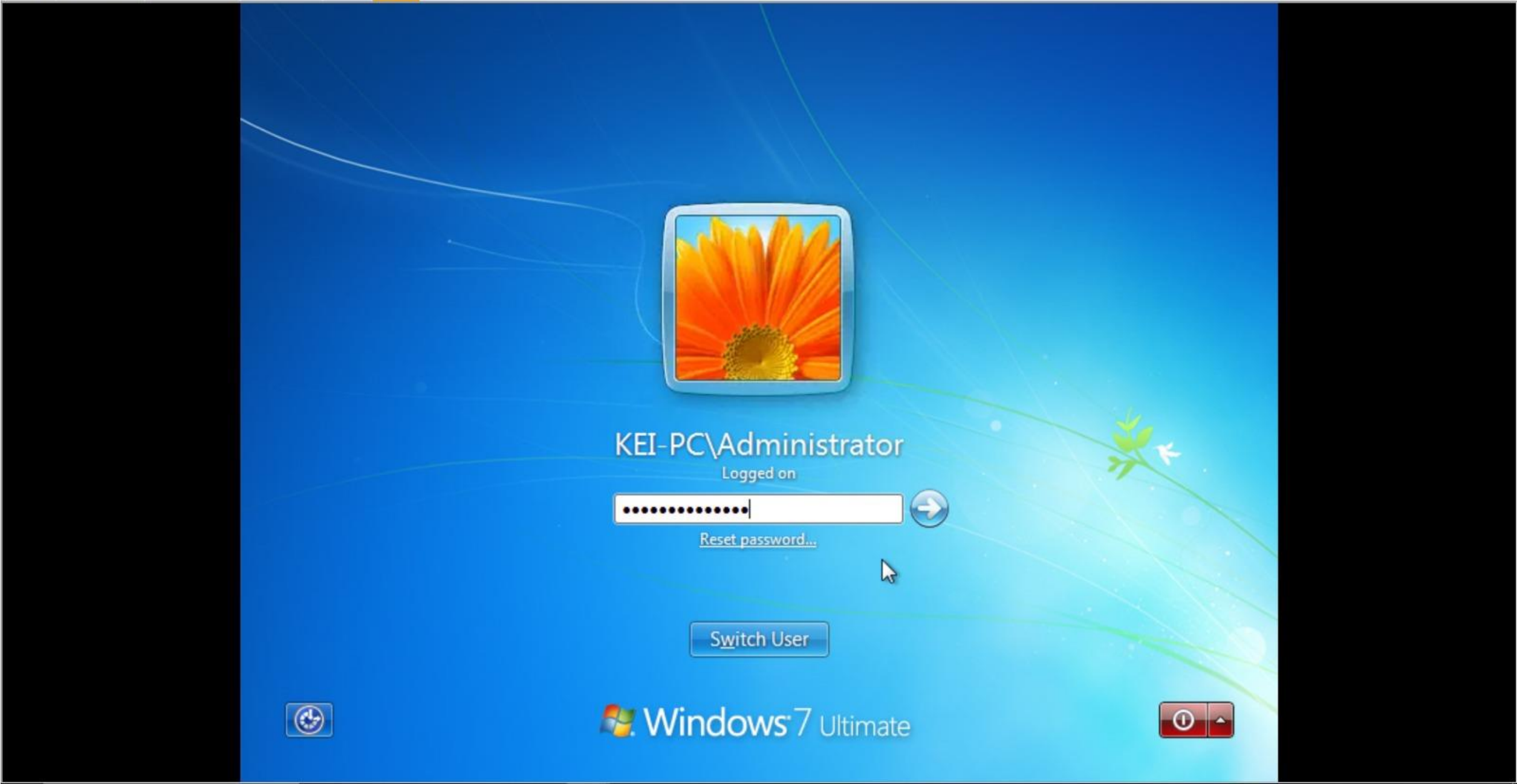
Computer name:

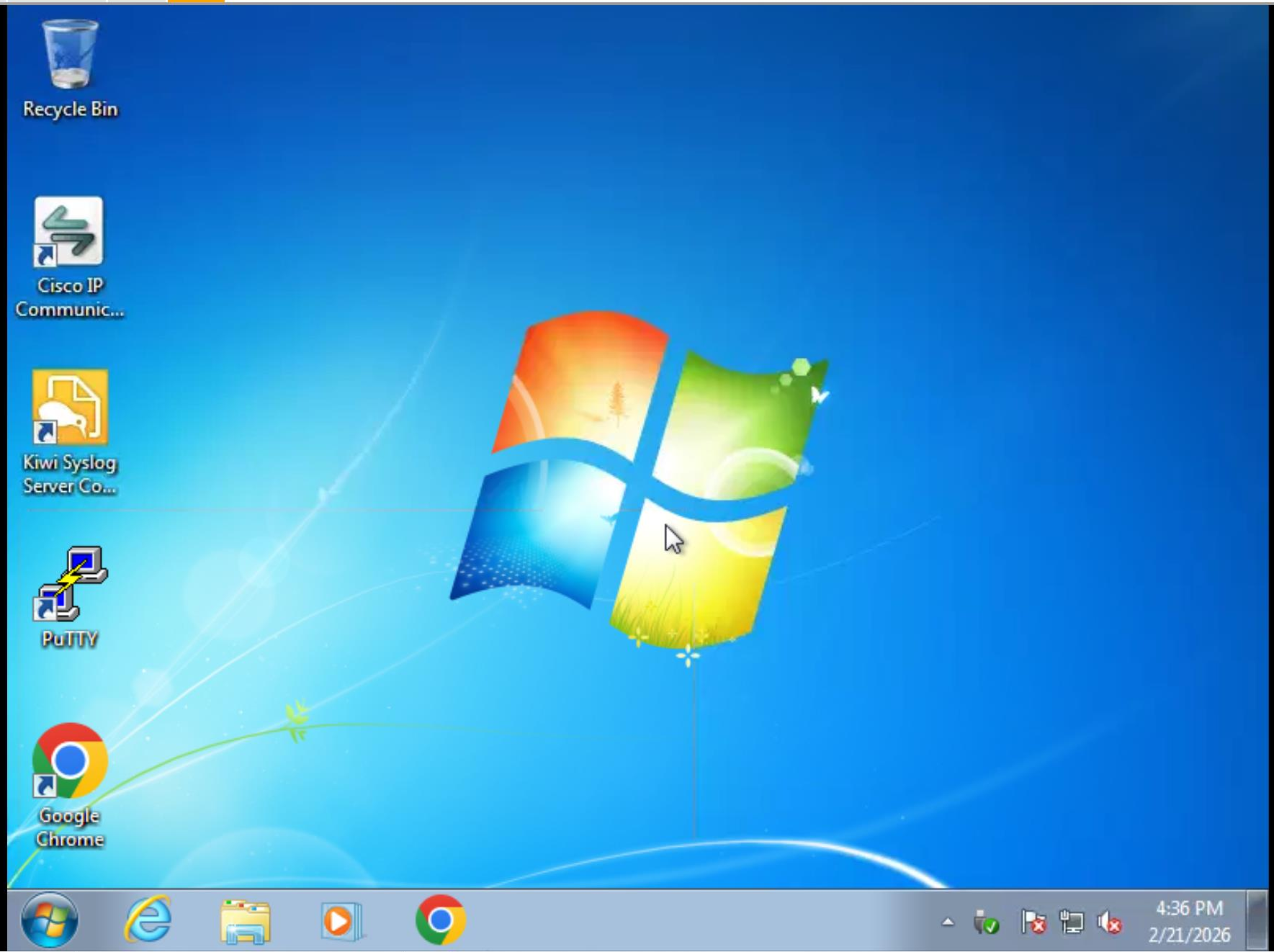
Password:

Password expires:

New expiration time (leave as is for immediate expiration):
 







password policy

The screenshot shows the 'Computer Configuration' tree on the left, expanded to 'Security Settings' > 'Account Policies' > 'Password Policies'. The 'Minimum password length' policy is selected and highlighted in blue. The right pane shows the policy settings:

Policy	Policy Setting
Enforce password history	24 passwords remembered
Maximum password age	42 days
Minimum password age	1 days
Minimum password length	12 characters
Password must meet complexity requirements	Enabled
Store passwords using reversible encryption	Disabled

Active Directory Users and Computers

File Action View Help

Group Policy Management Editor

File Action View Help

Computer Configuration

- Policies
 - Software Settings
 - Windows Settings
 - Name Resolut
 - Scripts (Startu
 - Security Settin
 - Account P
 - Passwe
 - Accou
 - Kerber
 - Local Polic
 - Event Log
 - Restricted
 - System Se
 - Registry
 - File System
 - Wired Net
 - Windows I
 - Network L
 - Wireless N
 - Public Key
 - Software F
 - Network A

Policy	Policy Setting
Account lockout duration	15 minutes
Account lockout threshold	5 invalid logon attempts
Reset account lockout counter after	15 minutes

Fine-Grained Password Policy

Create Password Settings: 100_days

TASKS ▼

SECTIONS ▼

Password Settings

Directly Applies To

Password Settings

Name: * 100_days

Precedence: * 1

☒ Enforce minimum password length

Minimum password length (characters): * 7

☒ Enforce password history

Number of passwords remembered: * 10

☒ Password must meet complexity requirements☐ Store password using reversible encryption☒ Protect from accidental deletion

Description:

Password age options:

☒ Enforce minimum password age

User cannot change the password withi... * 1

☒ Enforce maximum password age

User must change the password after (... * 100

☒ Enforce account lockout policy:

Number of failed logon attempts allowed: * 10

Reset failed logon attempts count after (m... * 60

Account will be locked out

Lockout duration (minutes): * 30

It is recommended to leave this option unchecked for better security protection.

Directly Applies To

Name

Mail

IT_HELPDESK

Add...

Remove

More Information

OK

Cancel



Active Directory...



Overview

KEI (local)

...\\Password Settings Contai...

System

Dynamic Access Control

Authentication

Authentication Policies

Global Search

Password Settings Container (1)

Filter



Name	Precedence	Type	Description
------	------------	------	-------------

100_days	1	Password S...	
----------	---	---------------	--

Tasks



Password Settings Container

New

Delete

Search under this node

Properties

WINDOWS POWERSHELL HISTORY



Loopback

Configure user Group Policy loopback processing mode

Configure user Group Policy loopback processing mode Previous Setting Next Setting

☐ Not Configured Comment:

☒ Enabled

☐ Disabled

Supported on: At least Windows 2000

Options: Mode: Merge

Help:

This policy setting directs the system to apply the set of Group Policy objects for the computer to any user who logs on to a computer affected by this setting. It is intended for special-use computers, such as those in public places, laboratories, and classrooms, where you must modify the user setting based on the computer that is being used.

By default, the user's Group Policy Objects determine which user settings apply. If this setting is enabled, then, when a user logs on to this computer, the computer's Group Policy Objects determine which set of Group Policy Objects applies.

If you enable this setting, you can select one of the following modes from the Mode box:

"Replace" indicates that the user settings defined in the computer's Group Policy Objects replace the user settings normally applied to the user.

"Merge" indicates that the user settings defined in the computer's Group Policy Objects and the user settings normally

OK Cancel Apply

OAC optimization
Policy
processing
line policy refresh

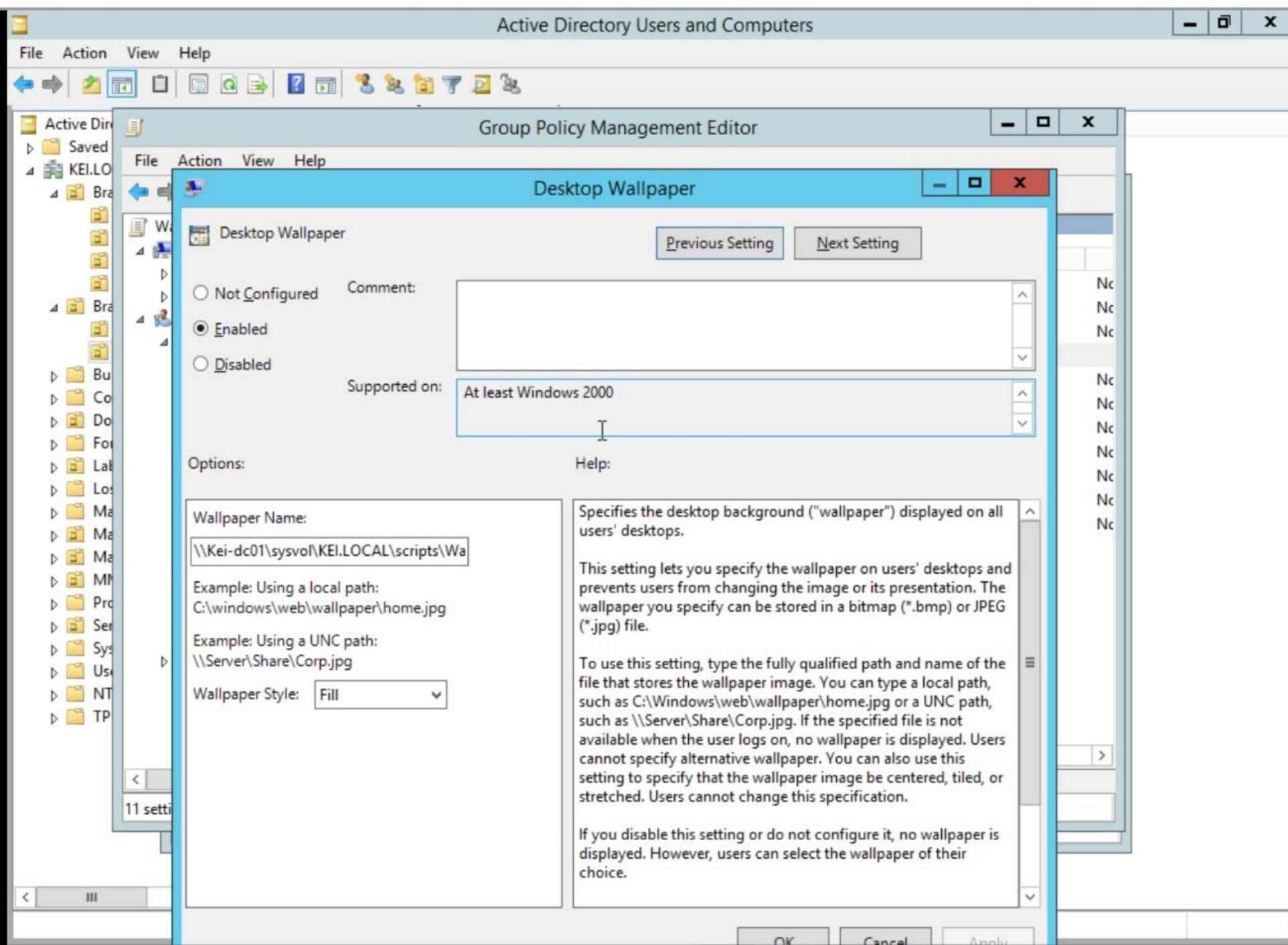
ers
ction
computers
omain controllers

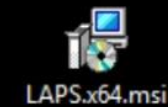
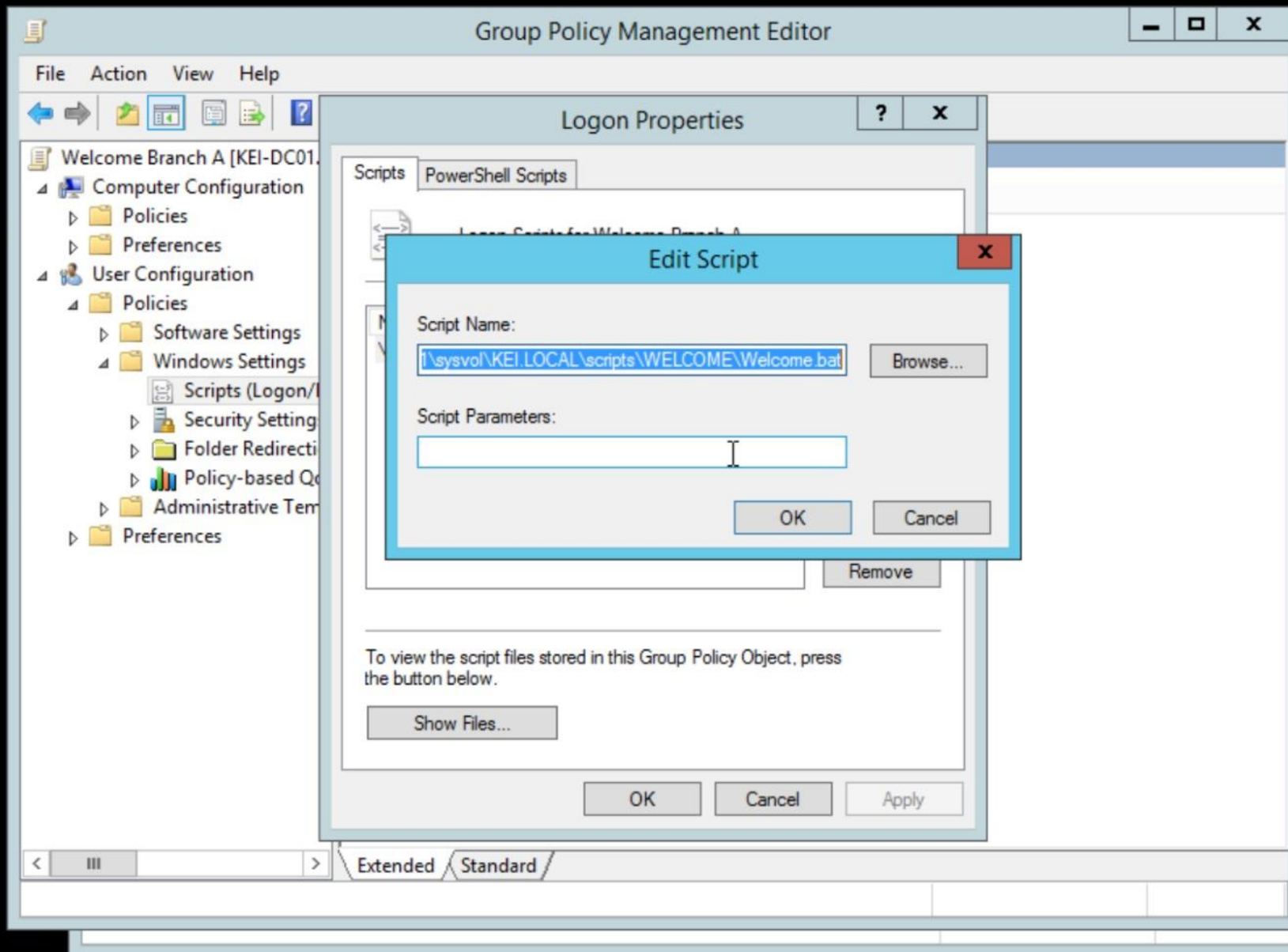
Policy Object Editor

g
chronization during p...
as a fast network conn...
in asynchronously wh...
time

k processing mode
y processing when log...
ension policy processi...

Desktop
Wallpaper
Policy/Logon
script







Recycle Bin



VLC media
player



Cisco IP
Communic...



WinSCP



Kiwi Sys-Log
Server Co...



Google
Chrome



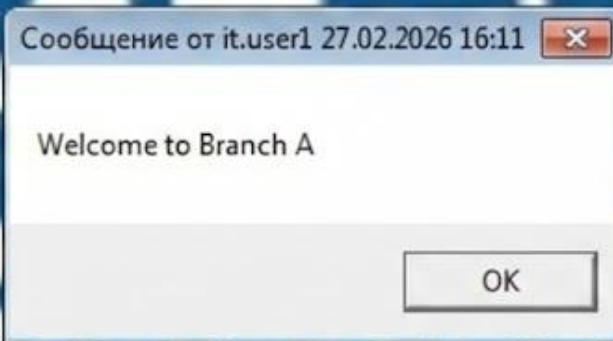
ManageEng...
Endpoint C...



LocalOffice...



PoliTV



4:11 PM
2/27/2026

Recycle Bin

Active Directory Administrative Center

Active Directory Administrative Center ▶ KEI (local) ▶

Manage Help

Active Directory... < KEI (local) (20)

Filter

Name	Type	Description
Branch A	Organizati...	
Branch B	Organizati...	
Builtin	builtinDom...	
Computers	Container	Default container for upgr...
Deleted Objects	Container	Default container for delet...
Domain Controllers	Organizati...	Default container for dom...
ForeignSecurityPrincipals	Container	Default container for secur...
Infrastructure	infrastructu...	
LabUsers	Organizati...	
LostAndFound	lostAndFou...	Default container for orph...
Managed Service Accounts	Container	Default container for man...
MattermostGroups	Organizati...	
MattermostUsers	Organizati...	

Branch A

Managed by: Address:

Modified: 2/18/2026 5:45 PM

Description:

Summary

Tasks

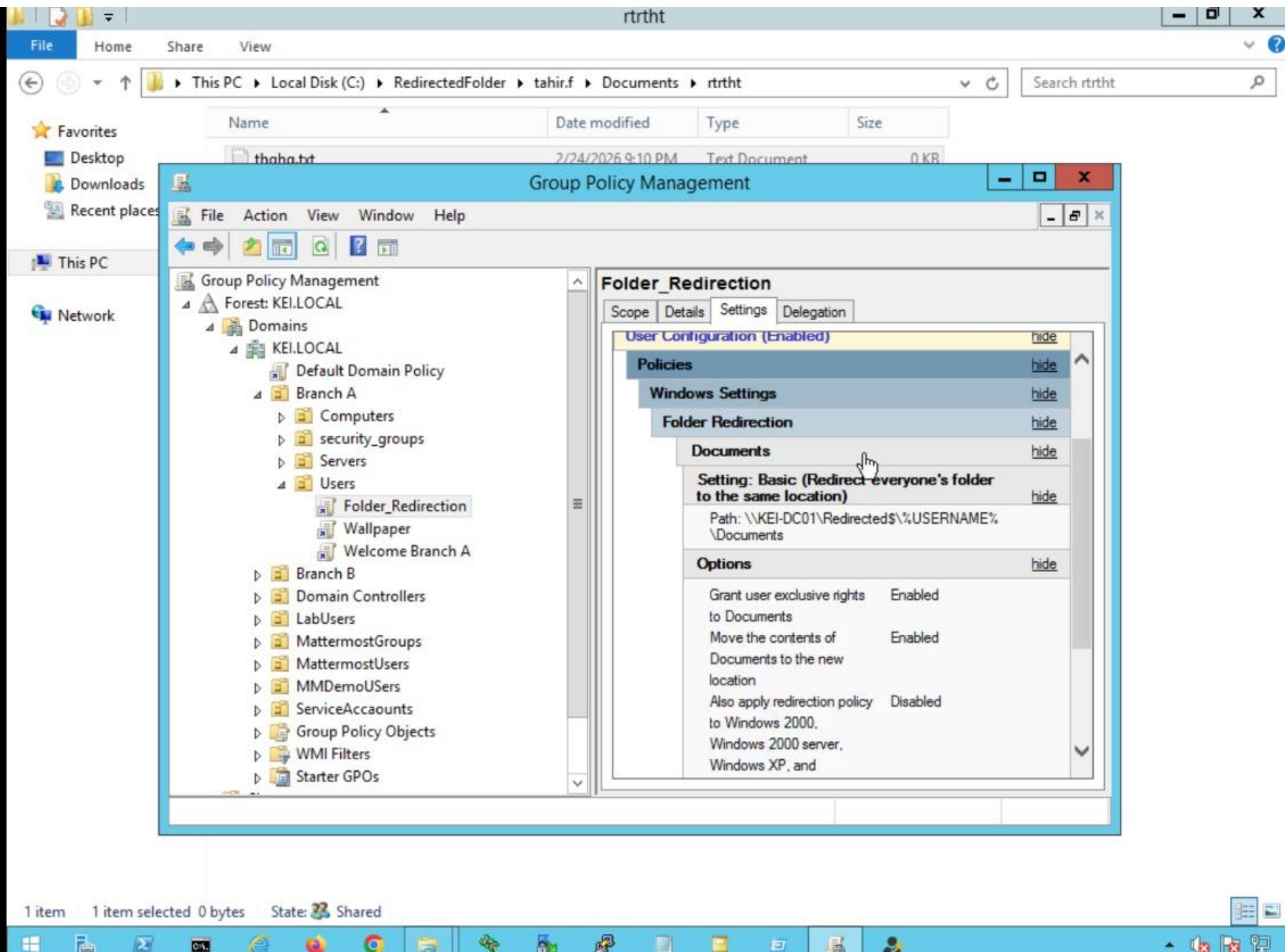
Branch A

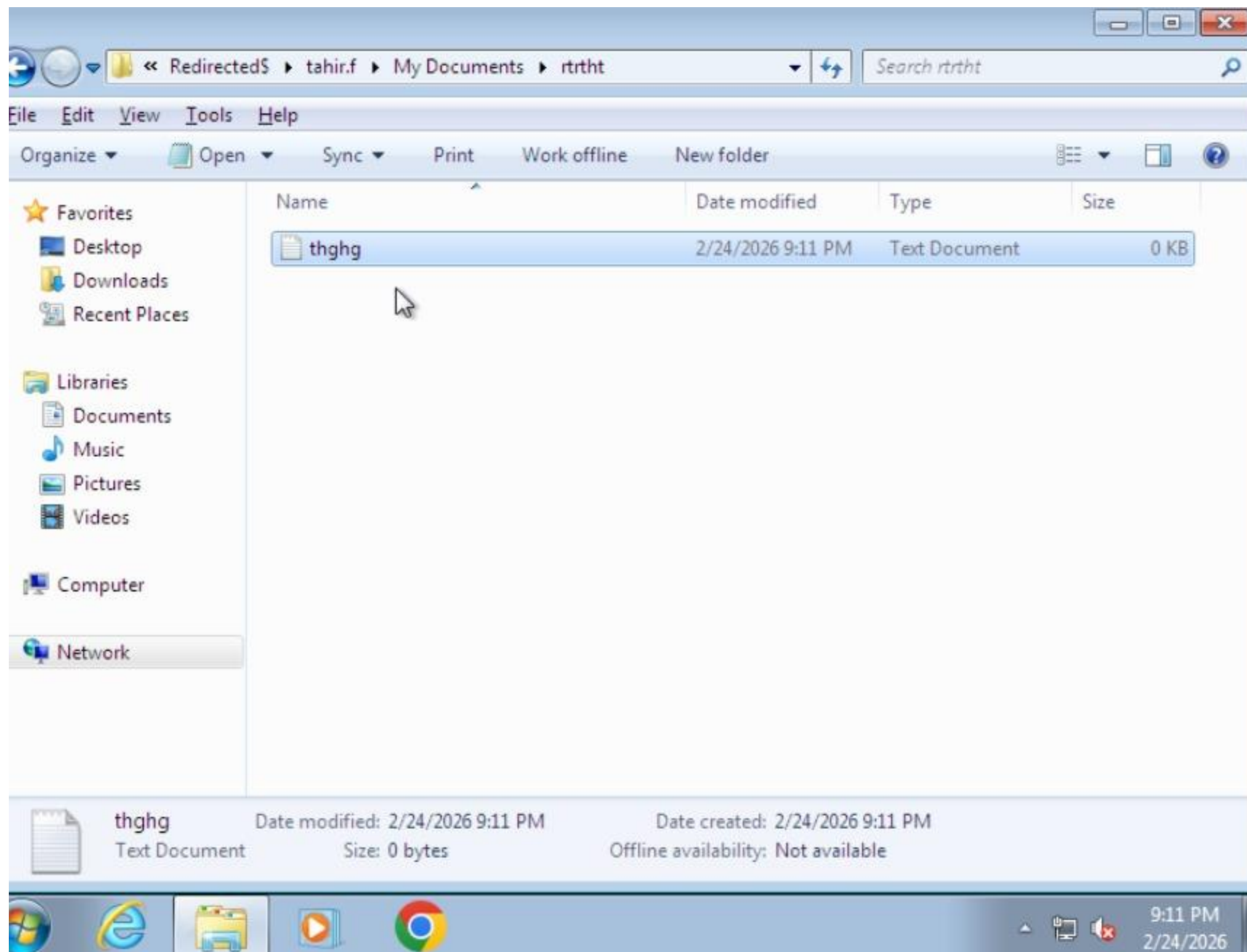
- New
- Delete
- Move...
- Search under this node
- Properties

KEI (local)

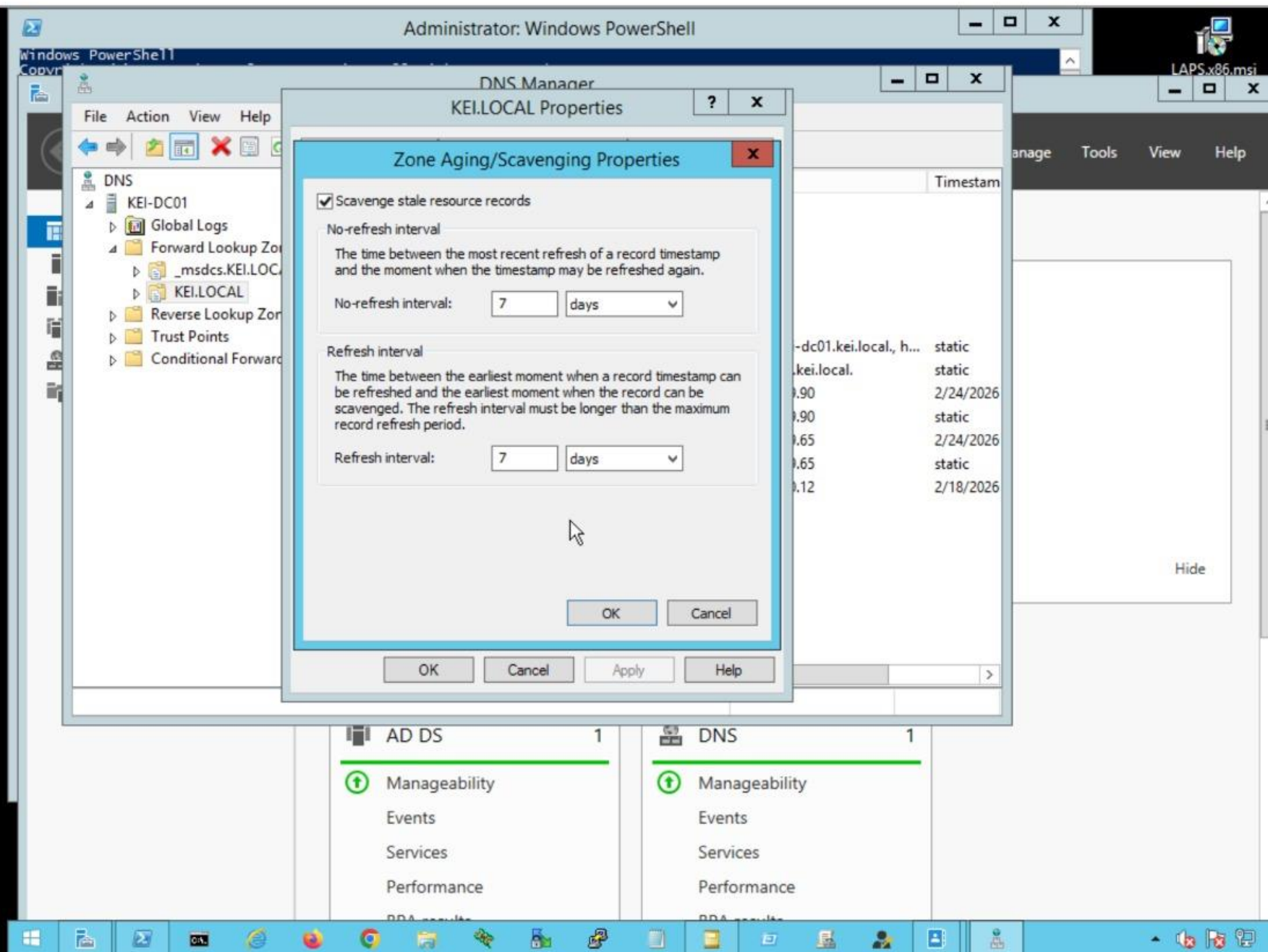
- Change domain controller
- Raise the forest functional leve...
- Raise the domain functional le...
- Enable Recycle Bin ...
- New
- Search under this node
- Properties

Folder Redirection





Aging and Scavenging



Integration of Linux systems with Active Directory domain

Feb 14 20:34

🔍 🔊 🔌

🔍

kamran@kamran-VMware-Virtual-Platform: ~/Desktop

🔍 = - □ ×

🔥

📁

🍷

?

>

📀

📀

📄

🗑️

🔄

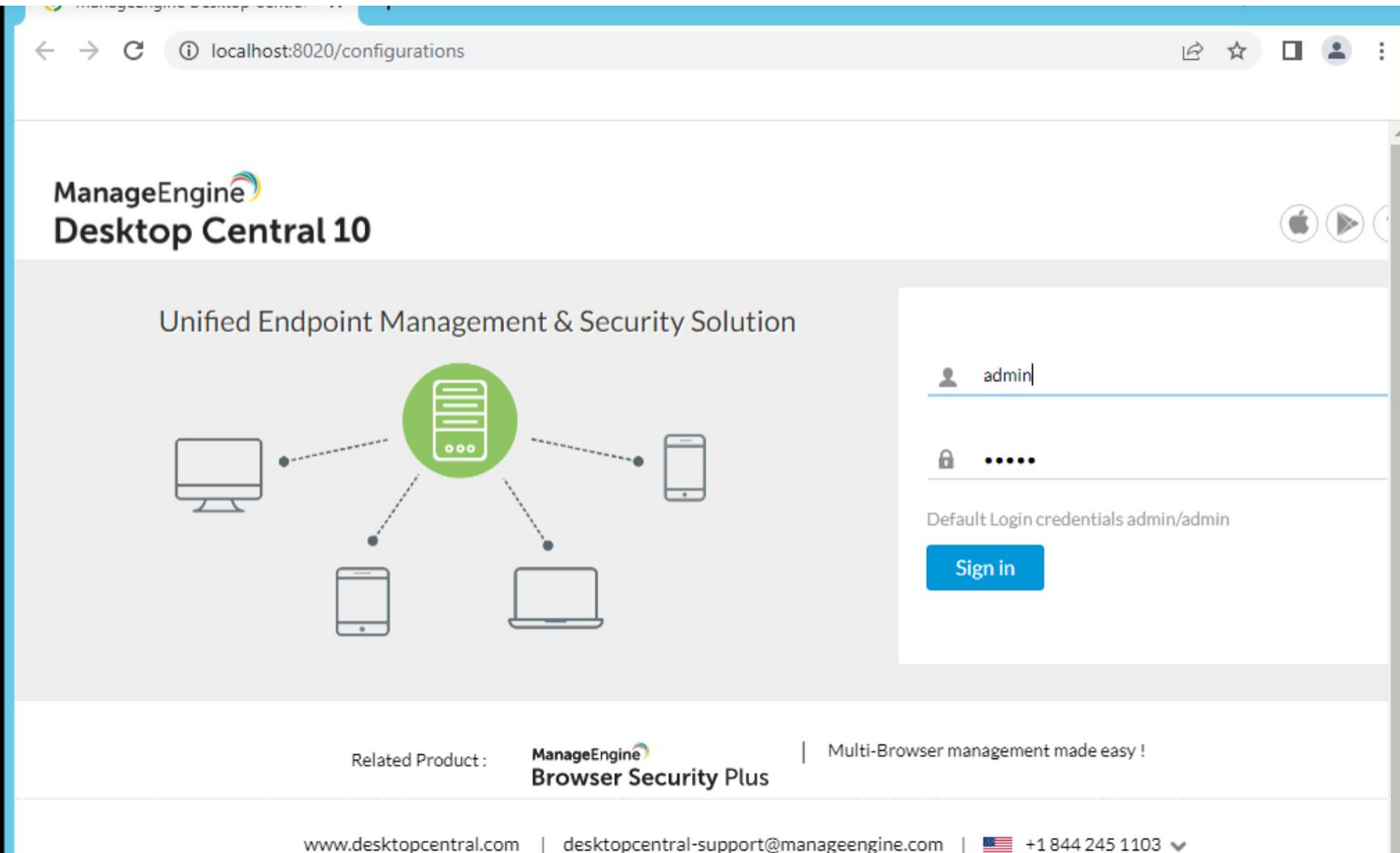
```
realm: No such realm found
kamran@kamran-VMware-Virtual-Platform:~/Desktop$ sudo nano /etc/resolv.conf
kamran@kamran-VMware-Virtual-Platform:~/Desktop$ sudo nano /etc/resolv.conf
kamran@kamran-VMware-Virtual-Platform:~/Desktop$ sudo nano /etc/resolv.conf
kamran@kamran-VMware-Virtual-Platform:~/Desktop$ sudo realm join KEI.LOCAL -U Administrator
Password for Administrator:
kamran@kamran-VMware-Virtual-Platform:~/Desktop$ realm list
KEI.LOCAL
type: kerberos
realm-name: KEI.LOCAL
domain-name: kei.local
configured: kerberos-member
server-software: active-directory
client-software: sssd
required-package: sssd-tools
required-package: sssd
required-package: libnss-sss
required-package: libpam-sss
required-package: adcli
required-package: samba-common-bin
login-formats: %U@kei.local
login-policy: allow-realm-logins
kamran@kamran-VMware-Virtual-Platform:~/Desktop$
```

🏠

Home

Desktop Central

The figure shows the login window for ManageEngine Desktop Central 10, a platform for centralized management of computers and devices on a network.



localhost:8020/webclient#/uems x +

localhost:8020/webclient#/uems/admin/database-backup

Home Configurations Patch Mgmt Sof

Global Settings

User Administration

Custom Group

Domain

Rebranding

Mobile App

Self Service Portal Settings

Credential Manager

Custom Field

SAML Authentication

SoM Settings

Scope of Management

Agent Settings

IP Scope

✓ Data base backup settings have been saved successfully

Exclude Desktop Central folder from Anti-virus scan:
Anti-Virus might restrict the Central server from synchronizing/downloading the Patch binaries, affect scheduled Database backup, agent upgradation, or upgrading Central server to latest version. [Learn More](#)

Want to migrate to Azure SQL database?

Schedule Database Backup

Everyday at * 04:01:00 [24 hour format]

Select ▼

Maintain last 7 backups only

Backup directory * C:\Program Files\DesktopCentral_Server\Sche

It is recommended to store your database back-up at a remote location to seamlessly restore the back-up if the Central Server fails.

☐ Notify when the database backup fails

☐ Secure the backup by setting password

Save Changes Cancel

Best Practices

12:05 PM

localhost:8020/webclient#/uems/ x +

localhost:8020/webclient#/uems/agent/domains

Home Configurations Patch Mgmt Software Deployment Inventory Agent Admin Support

Scope of Management

- Summary
- Domain
- Remote Offices
- Computers

Deployment

- SoM Policy
- Deploying Agent with OS

Settings

- Agent Settings
- IP Scope
- Retry Agent Installation
- Replication Policy

Add Domain

Domain Name * KEI.LOCAL

Network Type ☒ Active Directory ☐ WorkGroup

Domain User Name * Administrator

Password * ••••••••

AD Domain Name * KEI.LOCAL

Domain Controller Name * 192.168.0.11d

☐ Use LDAP SSL ?

Add Domain Cancel

(C) Copyright 2021, ZOHO Corp.

0 - 0 of 0 25

12:10 PM

ManageEngine Endpoint Centre X

https://kei.local:8383/webclient#/uems/agent/som/computers

Home Configurations Patch Mgmt Software Deployment Inventory OS Deployment Agent

Scope of Management

- Summary
- Domain
- Remote Offices
- Computers
- Prerequisites

Deployment

- SoM Policy
- Agent Installation
- Deploying Agent with OS

Settings

- Agent Settings
- IP Scope
- SoM Settings
- Replication Policy

Windows

Download Agent

+ Add Computer(s) X Remove Computer(s) Import Edit Credential

Want to customize computer details?

Agent Actions: Install Agent Uninstall Agent Move To Select Remote Office

Filters

Total Records | Search Grid Calendar Export

☐	Computer Name	Domain Name	Agent Status	Last Contact Time	Agent Version
☐	KEI-PC	KEI	Installed Succe...	Feb 22, 2026 08:20 ...	11.2.2300.2.W

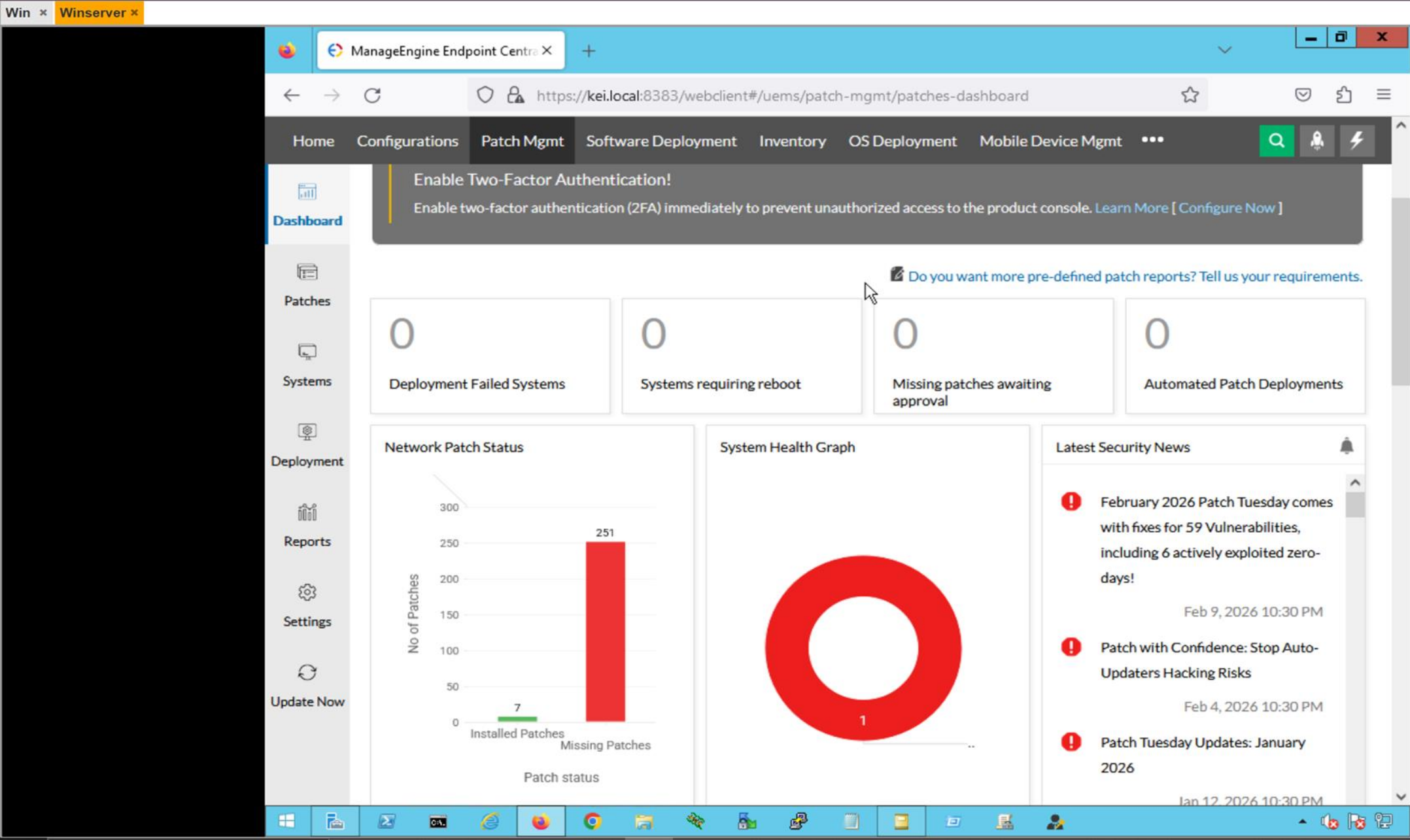
1 - 1 of Total Records 25

Quick Links Hide

How Tos Knowledge Base FAQ Videos

1. How to set admin credentials to run Central Server in Windows Vista and above?
2. How to install agents using Windows GPO/Startup script?
3. What are the different ways to install the agent in client computers?
4. How to install agents in remote office computers?
5. How to move Central Server from one machine to another machine?
6. How do I make users uninstall Agent from Add/Remove Programs?

The Patch Management control panel in Endpoint Central shows the status of patches, missing updates, installation errors, and system security.



The vulnerability and patch database is being updated — the system synchronizes update information with the ManageEngine server.

WinserverWin

ManageEngine DesktopCentral 10

Jump to SDPLicenseVersion : 10.0.656

HomeConfigurationsPatch MgmtSoftware DeploymentInventoryAgentAdminSupport

SearchUserRefresh

DashboardPatchesSystemsDeploymentReportsSettingsUpdate Now

Vulnerability Database Update is in Progress

Patch information from the online Central Repository Server is being synchronized

Note: Patch Management related tasks cannot be performed while sync is in progress.You will be automatically redirected after the synchronization is completed.

Sync Type: Manual

Sync Start Time: Feb 15, 2026 09:33 AM

Sync-Initiated By: admin

Process	Status
Connecting To The Central Repository	Completed
Downloading And Syncing Windows Patch SQLs	Completed
Downloading And Syncing Mac Patch SQLs	Completed
Checking For New Updates	Completed
Downloading And Syncing Ubuntu Patch SQLs	In progress

ManageEngine Desktop Central x

ManageEngine Desktop Central x

ManageEngine Desktop Central x

+

← → ↻ ⚠ Not secure | https://localhost:8383/webclient#/uems/admin/custom-group/320

Home Configurations Patch Mgmt Software Deployment Inventory Agent Admin Support ...

Global Settings

User Administration

Custom Group

Domain

Rebranding

Mobile App

Self Service Portal Settings

Credential Manager

Custom Field

SAML Authentication

SoM Settings

Scope of Management

Agent Settings

IP Scope

Custom Group > View Group

View Group (Test-PC)

Custom Group

Edit Delete

Group Name : Test-PC

Created By : admin

Group Type : Computer

Created Time : Feb 15, 2026 09:45 AM

Group Category : Static

Last Modified By : admin

Last Modified Time : Feb 15, 2026 09:45 AM

Selected Domain : KEI

Members Count : 1

Total : 1

Q

Table View

Download

☐	Computer Name	Domain Name	OS Name	Last Contact Time	Remote Office
☐	KEI-CLIENT02	KEI	Windows 7 UL...	Feb 15, 2026 09:22 ...	Local Office

1 - 1 of 1

25

Windows Taskbar

1:45 PM

A task has been created to automatically install updates (patches) — the system will download and install updates on a schedule.

Winserver xWin x

ManageEngine Desktop Central

Automate Patch Deployment task MyTask1 has been created successfully

Version : 10.0.656

HomeConfigurationsPatch MgmtSoftware DeploymentInventoryAgentAdminSupport

Search

Dashboard

Patches

Systems

Deployment

Reports

Settings

Update Now

Deployment

Deployment Policies

Trash

Collapse Sidebar

How Automated Patch Deployment works?

Show Tasks

Created by AllCreated by Me

+ Automate Task

Actions

Filter By : Platform

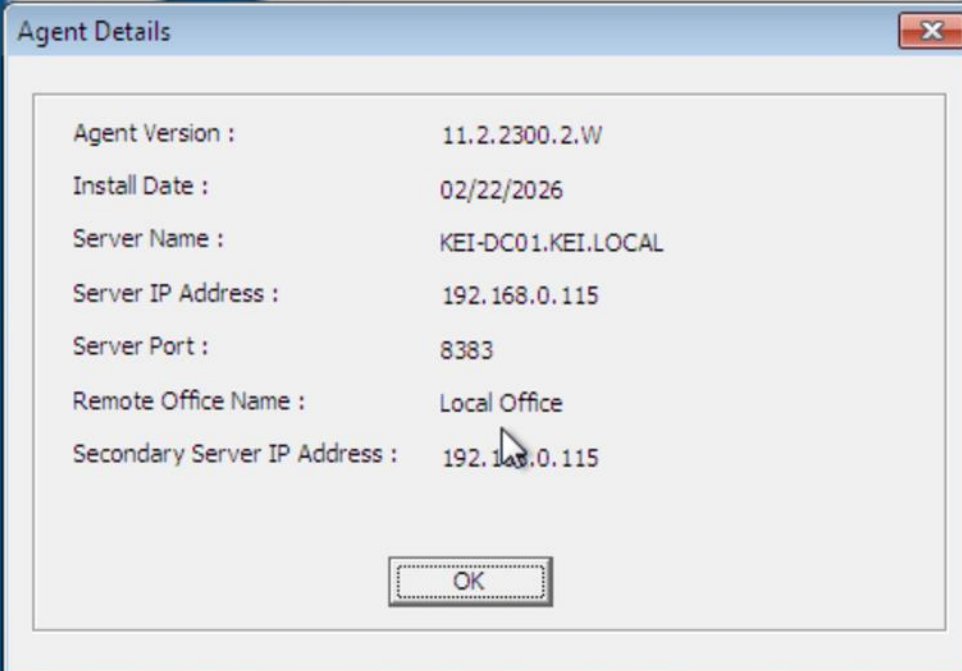
Total : 1

	Name	Deployment Time	Created Time	Current Status	Deployment Option	Migrati
	MyTa...	00:00 to 23:59	Feb 15, 2026 10...	1	Download and Dep...	--

1 - 1 of 125

Activate Windows

Go to System in Control Panel to activate Windows.



Win × Winserver ×

Win × Winserver ×

Packages

Templates

Install/Uninstall Software >

[View Configurations](#)

User-defined Templates

Trash

Self Service Portal

Self Service Portal Reports

SSP Usage Reports

Software Repository

Script Repository

Auto-update Templates

Proxy Settings

Deployment Policies

Unpublished packages in SSP

1

Packages in SSP which are not in use

0

[▶ Watch Demo](#)

Need to deploy software to other Linux OS's?

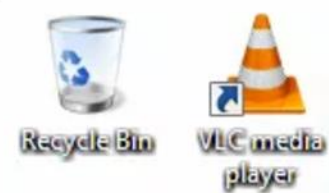
Filter By: Package Type License Type Platform

[+ Add Package](#)
[x Delete Packages](#)
[Install/Uninstall Software](#)
[Mark License Type](#)

Total: 1 | ✕ 📊 📄

☒	Package Name	Software Name	Created By	Modified Time	Auto-update status	Platform	Action
	7-zip	String Only					
☒	7-zip	--	admin	Feb 22, 2026 12:5...	Not Applicable	Windows	

1 - 1 of 1



ManageEngine Endpoint Central 5.5 Self Service Portal

Apps (4) Updates (0)

Search: All Status All Apps Published time: All Last Sync: 02/22/2026 01:04:18 PM

Application Name	Version	Vendor	Description	Published Time	Action	Install
1.1 Trup				Feb 22, 2026 12:57 PM	Installing	



Wazuh

(Security / SIEM)

< Deploy new agent

✓ Select the package to download and install on your system:



LINUX

- ☐ RPM amd64 ☐ RPM aarch64
- ☐ DEB amd64 ☐ DEB aarch64



WINDOWS

- ☒ MSI 32/64 bits



macOS

- ☐ Intel
- ☐ Apple silicon

④ For additional systems and architectures, please check our [documentation](#).

✓ Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address ?

172.20.28.178

☐ Remember server address

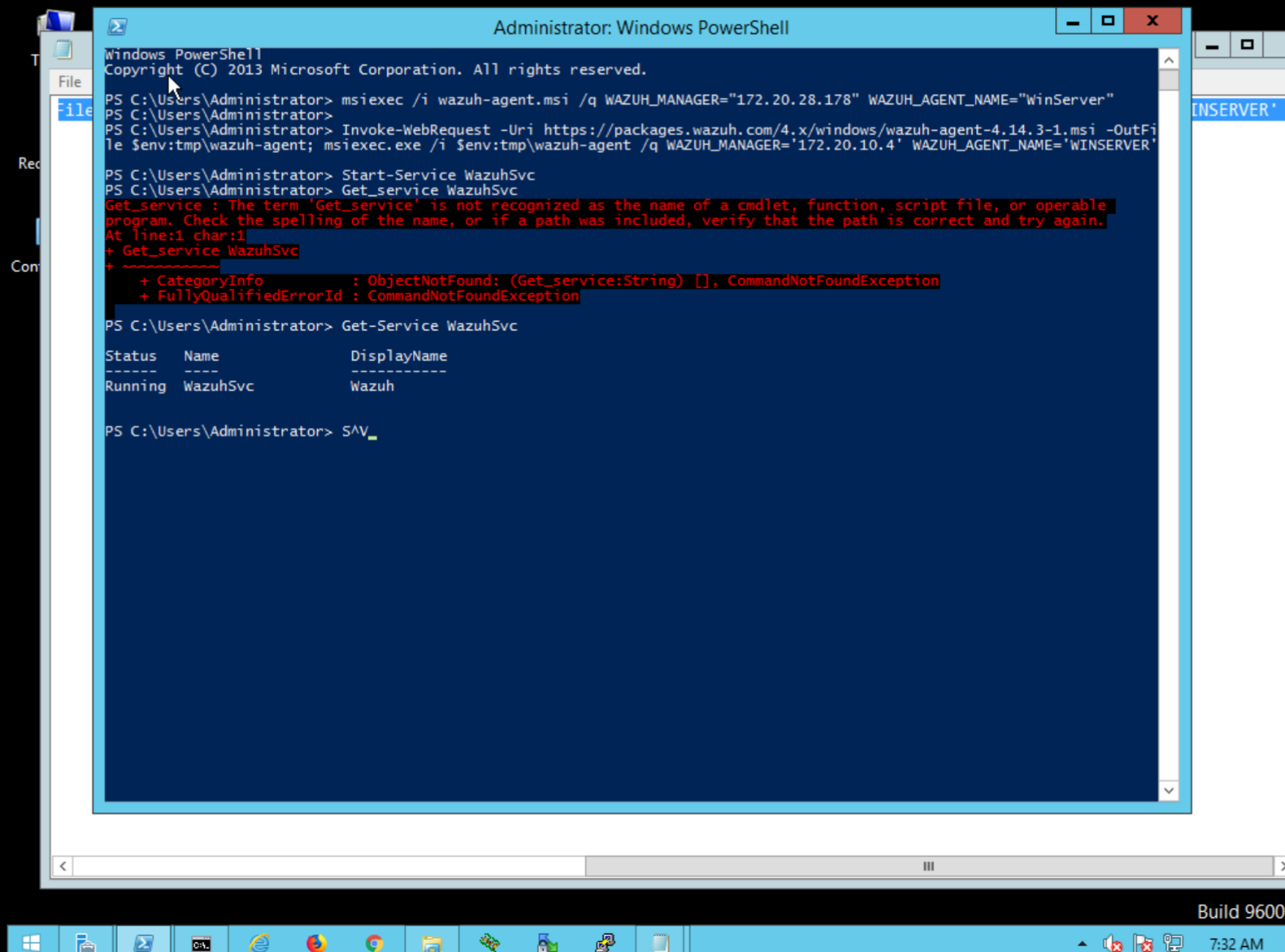
✓ Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name: ?

WinServer

④ The agent name must be unique. It can't be changed once the agent has been enrolled.



The screenshot shows a Windows desktop environment. A Windows PowerShell window titled "Administrator: Windows PowerShell" is open, displaying a series of commands and their outputs. The commands include installing the Wazuh agent, making a web request, starting the Wazuh service, and checking its status. An error message is shown for the `Get-service` command, which is highlighted in red. The taskbar at the bottom shows various application icons, including File Explorer, PowerShell, and web browsers. The system tray in the bottom right corner displays the time as 7:32 AM and the system build number as Build 9600.

```
Administrator: Windows PowerShell

Windows PowerShell
Copyright (C) 2013 Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> msixec /i wazuh-agent.msi /q WAZUH_MANAGER="172.20.28.178" WAZUH_AGENT_NAME="WinServer"
PS C:\Users\Administrator>
PS C:\Users\Administrator> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.3-1.msi -OutFi
le $env:tmp\wazuh-agent; msixec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='172.20.10.4' WAZUH_AGENT_NAME='WINSERVER'

PS C:\Users\Administrator> Start-Service WazuhSvc
PS C:\Users\Administrator> Get-service WazuhSvc
Get-service : The term 'Get-service' is not recognized as the name of a cmdlet, function, script file, or operable
program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ Get-service WazuhSvc
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (Get-service:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Administrator> Get-Service WazuhSvc

Status      Name      DisplayName
-----
Running     WazuhSvc  Wazuh

PS C:\Users\Administrator> SAV_
```

Build 9600 7:32 AM



AGENTS BY STATUS



- Active (2)
- Disconnected (1)
- Pending (0)
- Never connected (0)

TOP 5 OS



- windows (2)
- ubuntu (1)

TOP 5 GROUPS



- default (3)

Agents (2)

[Deploy new agent](#) [Refresh](#) [Export formatted](#) [More](#) [Settings](#)

status=active

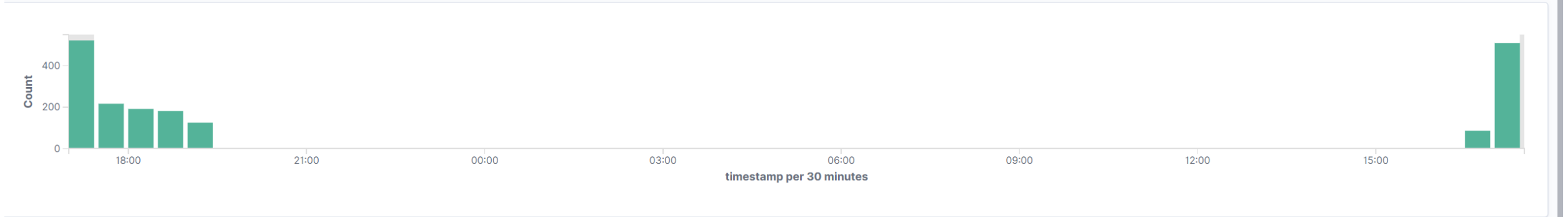
WQL

☐	ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
☐	001	WINSERVER	172.20.29.90	default	Microsoft Windows Server 2012 R2 Standard 6.3.9600	node01	v4.14.3	● active ⓘ	
☐	003	Ubuntu	172.20.29.120	default	Ubuntu 24.04.4 LTS	node01	v4.14.3	● active ⓘ	

Rows per page: 10

manager.name: wazuh-server

🔍 + Add filter



1,835 hits

Feb 20, 2026 @ 17:25:19.585 - Feb 21, 2026 @ 17:25:19.585

📄 Export Formatted

🔄 Reset view

🔍 707 available fields

📄 Columns

📄 Density

📄 1 fields sorted

🖥 Full screen

	timestamp	agent.name	rule.description	rule.level	rule.id
🔍	Feb 21, 2026 @ 17:25:11.0...	WINSERVER	Windows User Logoff	3	60137
🔍	Feb 21, 2026 @ 17:25:11.0...	WINSERVER	Windows Logon Success	3	60106
🔍	Feb 21, 2026 @ 17:25:11.0...	WINSERVER	Windows User Logoff	3	60137
🔍	Feb 21, 2026 @ 17:25:11.0...	WINSERVER	Windows Logon Success	3	60106
🔍	Feb 21, 2026 @ 17:25:08.1...	WINSERVER	Windows User Logoff	3	60137
🔍	Feb 21, 2026 @ 17:25:08.1...	WINSERVER	Windows Logon Success	3	60106
🔍	Feb 21, 2026 @ 17:25:07.7...	Ubuntu	PAM: Login session closed.	3	5502
🔍	Feb 21, 2026 @ 17:25:07.7...	Ubuntu	PAM: Login session opened.	3	5501
🔍	Feb 21, 2026 @ 17:25:07.7...	Ubuntu	Successful sudo to ROOT executed.	3	5402
🔍	Feb 21, 2026 @ 17:24:31.0...	WINSERVER	Windows User Logoff	3	60137



Recycle Bin



VLC media
player



Cisco IP
Communic...



Kiwi Syslog
Server Co...



ManageEng...
Endpoint C...



PuTTY

CA: Administrator: C:\Windows\System32\cmd.exe

```
C:\Windows\system32>net user hacker P@ssw0rd123! /add  
The command completed successfully.
```

```
C:\Windows\system32>net localgroup administrator hacker /add  
System error 1376 has occurred.
```

The specified local group does not exist.

```
C:\Windows\system32>net localgroup administrators hacker /add  
The command completed successfully.
```

```
C:\Windows\system32>_
```

IT ACADEMY



W.

Threat Hunting



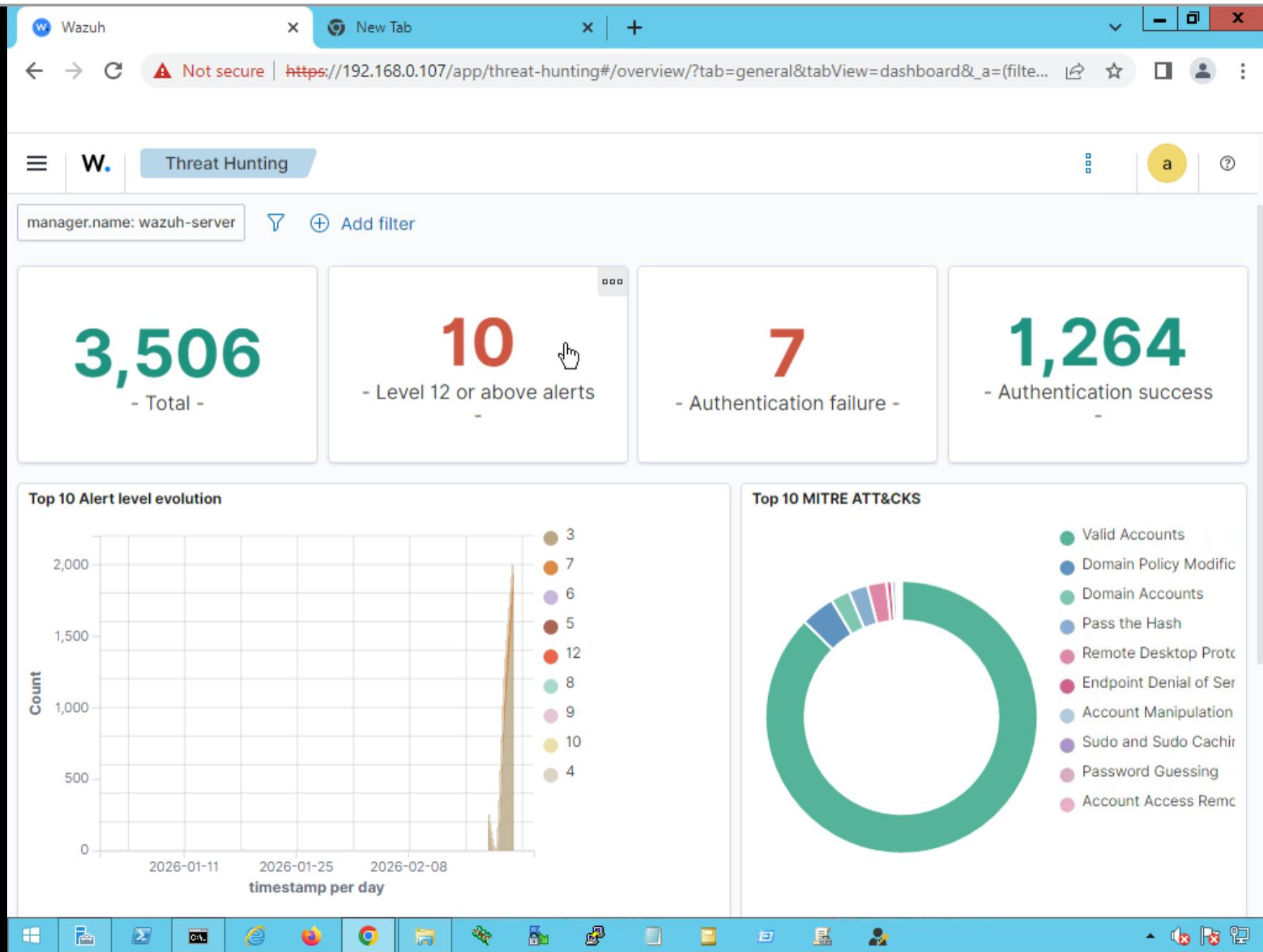
a

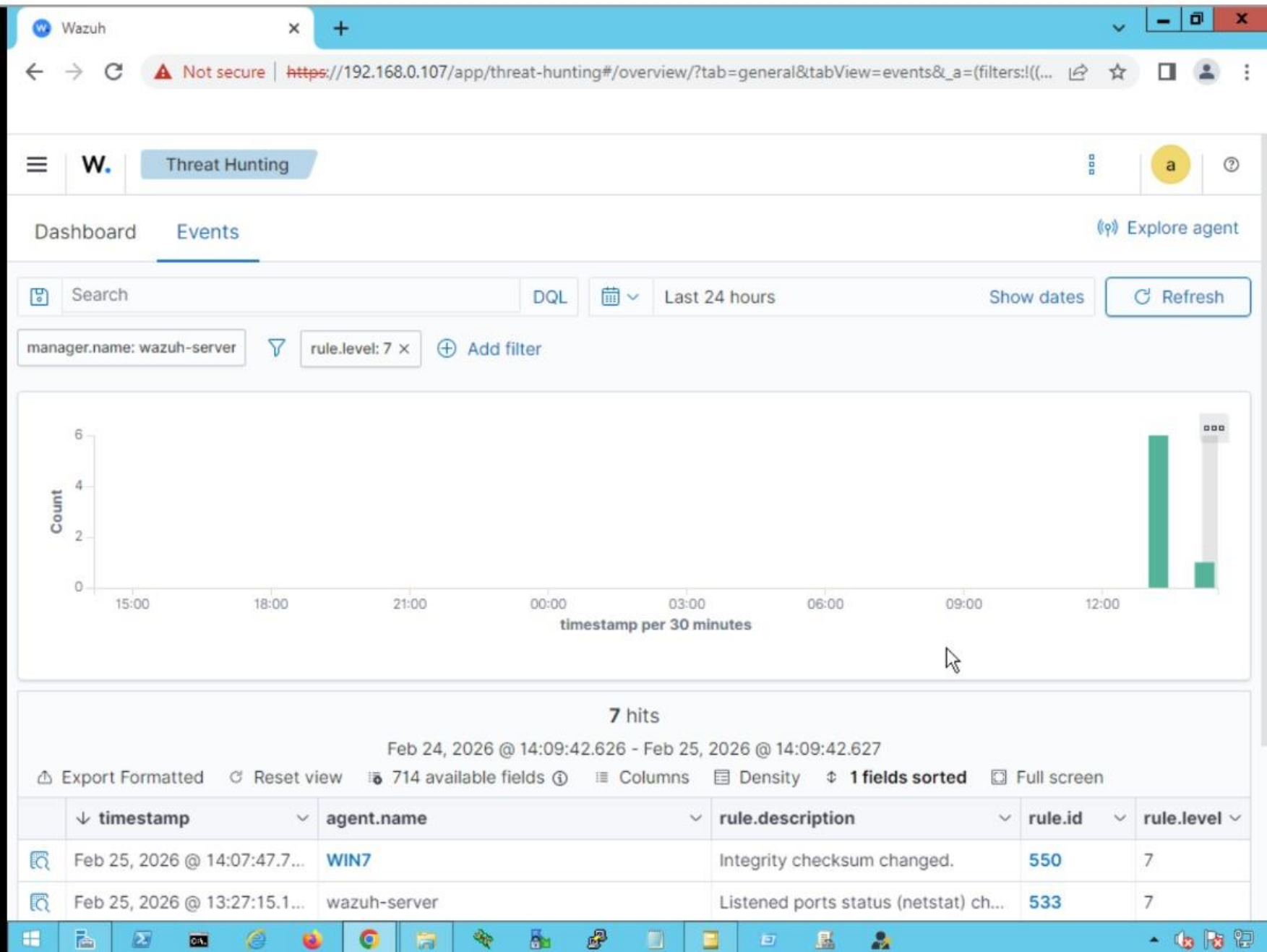


	↓ timestamp	agent.name	rule.description	rule.level	rule.id
	Feb 25, 2026 @ 13:34:30.0...	WIN7	Administrators Group Changed	12	60154
	Feb 25, 2026 @ 13:34:21.4...	WINSERVER	Windows User Logoff	3	60137
	Feb 25, 2026 @ 13:34:21.4...	WINSERVER	Windows Logon Success	3	60106
	Feb 25, 2026 @ 13:34:19.0...	WINSERVER	Windows User Logoff	3	60137
	Feb 25, 2026 @ 13:34:19.0...	WINSERVER	Windows User Logoff	3	60137
	Feb 25, 2026 @ 13:34:19.0...	WINSERVER	Windows User Logoff	3	60137
	Feb 25, 2026 @ 13:34:18.9...	WINSERVER	Windows Logon Success	3	60106
	Feb 25, 2026 @ 13:34:18.9...	WINSERVER	Windows Logon Success	3	60106
	Feb 25, 2026 @ 13:34:04.2...	WIN7	Users Group Changed	5	60170
	Feb 25, 2026 @ 13:34:04.2...	WIN7	User account changed	8	60110
	Feb 25, 2026 @ 13:34:04.1...	WIN7	User account enabled or created	8	60109
	Feb 25, 2026 @ 13:34:04.1...	WIN7	User account enabled or created	8	60109
	Feb 25, 2026 @ 13:34:04.1...	WIN7	Domain Users Group Changed	5	60160
	Feb 25, 2026 @ 13:33:58.7...	WINSERVER	Windows User Logoff	3	60137
	Feb 25, 2026 @ 13:33:58.6...	WINSERVER	Windows User Logoff	3	60137

Rows per page: 15

< 1 2 3 4 5 ... 9 >





Mattermost + LDAP

```
root@kamran-VMware-Virtual-Platform:/home/kamran/mattermost# ls
docker-compose.yml
root@kamran-VMware-Virtual-Platform:/home/kamran/mattermost# nano docker-compose.yml
root@kamran-VMware-Virtual-Platform:/home/kamran/mattermost# docker compose down -v
WARN[0000] /home/kamran/mattermost/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] down 4/4
✓ Container mm_app Removed 0.6s
✓ Container mm_postgres Removed 0.3s
✓ Volume mattermost_dbdata Removed 0.1s
✓ Network mattermost_default Removed 0.1s
root@kamran-VMware-Virtual-Platform:/home/kamran/mattermost# docker compose up -d
WARN[0000] /home/kamran/mattermost/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 21/21
✓ Image postgres:15 Pulled 16.6s
✓ Network mattermost_default Created 0.1s
✓ Volume mattermost_dbdata Created 0.3s
✓ Container mm_postgres Created 0.4s
✓ Container mm_app Created 0.1s
root@kamran-VMware-Virtual-Platform:/home/kamran/mattermost# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS      NAMES
d0cf2e8a3654   mattermost/mattermost-team-edition  "/mattermost/bin/mat...  4 seconds ago Up 3 second mm_app
s (health: starting) 8067/tcp, 0.0.0.0:8065->8065/tcp, [::]:8065->8065/tcp, 8074-8075/tcp
del44cfacc4f   postgres:15                          "docker-entrypoint.s...  4 seconds ago Up 3 second  mm_postgre
s
5432/tcp
root@kamran-VMware-Virtual-Platform:/home/kamran/mattermost# ls
docker-compose.yml
root@kamran-VMware-Virtual-Platform:/home/kamran/mattermost# ss -tlnp | grep 8065
LISTEN 0      4096      0.0.0.0:8065      0.0.0.0:*      users:({"docker-proxy",pid=13335,fd=8})

LISTEN 0      4096      [::]:8065        [::]:*        users:({"docker-proxy",pid=13340,fd=8})
```




What's the name of your organization?

We'll use this to help personalize your workspace.

Continue

KEI IT PROJECT ▾

+

☰

🔍 Find channel

Threads

CHANNELS

🌐

LAB-WORK

🌐

Off-Topic

🌐

Town Square

🔒

step

+

Add channels

DIRECT MESSAGES

+

Invite Members

☆ Town Square ▾ 👤 4 📄

Start call ⓘ



Town Square

Welcome to Town Square. Post messages here that you want everyone to see. Everyone automatically becomes a member of this channel when they join the team.

📧 Invite others to the workspace

Saturday

🔊

System 5:55 PM

You and [@kamran103](#) joined the team.

K

kamran103 5:57 PM

Salam!

I

inal 5:57 PM

SALAM!

Wednesday

🔊

System 7:31 PM

[@yusif.a](#) joined the team.

Y

yusif.a 7:31 PM

Salam!

🔊

System 7:31 PM

[@ali.m](#) joined the team.

A

ali.m 7:31 PM

Salam!

Write to Town Square

B

I

🔗

H

🔗

<>

🗨️

📋

📋

🕒

👤

Aa ^

📎

😊

➡

⌵

[← Back to kei-it-project](#)

System Console
@kamran103

Find settings

AD/LDAP

Plugins

Integrations

Compliance

AD/LDAP Wizard

SECTIONS

- Connection Settings
- User Filters
- Account sync
- Group Synchronization
- Sync Performance
- Sync History

Connection Settings

Connection and security level to your AD/LDAP server.

Enable sign-in with AD/LDAP: ☒ True ☐ False
When true, Mattermost allows login using AD/LDAP

Enable Synchronization with AD/LDAP: ☒ True ☐ False
When true, Mattermost periodically synchronizes users from AD/LDAP. When false, user attributes are updated from AD/LDAP during user login only.

Login Field Name:
The placeholder text that appears in the login field on the login page. Defaults to "AD/LDAP Username".

AD/LDAP Server:
The domain or IP address of AD/LDAP server.

AD/LDAP Port:
The port Mattermost will use to connect to the AD/LDAP server. Default is 389.

Bind Username:

Save

AD/LDAP Wizard

SECTIONS

Connection Settings

User Filters

Account sync

Group Synchronization

Sync Performance

Sync History

Connection Settings

Connection and security level to your AD/LDAP server.

Enable sign-in with AD/LDAP: ☒ True ☐ False

When true, Mattermost allows login using AD/LDAP

Enable Synchronization with AD/LDAP: ☒ True ☐ False

When true, Mattermost periodically synchronizes users from AD/LDAP. When false, user attributes are updated from AD/LDAP during user login only.

Login Field Name:

Domain Username

The placeholder text that appears in the login field on the login page. Defaults to "AD/LDAP Username".

AD/LDAP Server:

172.20.29.65

The domain or IP address of AD/LDAP server.

AD/LDAP Port:

389

The port Mattermost will use to connect to the AD/LDAP server. Default is 389.

Bind Username:

ldapbind@kei.local

The username used to perform the AD/LDAP search. [More Info](#)

Save

[← Back to kei-it-project](#)

System Console
@inal

☰

🔍 Find settings

🕒 ABOUT

📄 Edition and License

📊 REPORTING

📈 Workspace Optimization

📊 Site Statistics

📊 Team Statistics

📄 Server Logs

👤 USER MANAGEMENT

👤 Users

👤 Groups

👤 Teams

👤 Channels

👤 Permissions

📋 SYSTEM ATTRIBUTES

👤 User Attributes

👤 Attribute-Based Access

📁 ENVIRONMENT

🌐 Web Server

🗄 Database

🔔 5

AD/LDAP Wizard

📋 SECTIONS

🔗 Connection Settings

👤 User Filters

🔄 Account sync

🔄 Group Synchronization

📊 Sync Performance

📋 Sync History

Maximum Page Size:

0

The maximum number of users the Mattermost server will request from the AD/LDAP server at one time. 0 is unlimited.

Query Timeout (seconds):

60

The timeout value for queries to the AD/LDAP server. Increase if you are getting timeout errors caused by a slow AD/LDAP server.

Synchronize users to the system

See the table below for the status of each synchronization

AD/LDAP Synchronize Now

Initiates an AD/LDAP synchronization immediately. See the table below for status of each synchronization. Please review "System Console > Logs" and [documentation](#) to troubleshoot errors.

Status	Finish Time	Run Time	Details
Success	Feb 18, 2026 - 07:27 PM	1 seconds	Scanned 7 LDAP users and 49 groups.
Success	Feb 18, 2026 - 07:24 PM	1 seconds	Scanned 3 LDAP users and 0 groups.

Save

[← Back to kei-it-project](#)

System Console

@inal

Find settings

ABOUT

Edition and License

REPORTING

Workspace Optimization

Site Statistics

Team Statistics

Server Logs

USER MANAGEMENT

Users

Groups

Teams

Channels

Permissions

SYSTEM ATTRIBUTES

User Attributes

Attribute-Based Access

ENVIRONMENT

Web Server

Database

File Storage

5

Mattermost Users

Revoke All Sessions

Search users

Filters (0)

Columns
9 selected

Duration
All time

Export

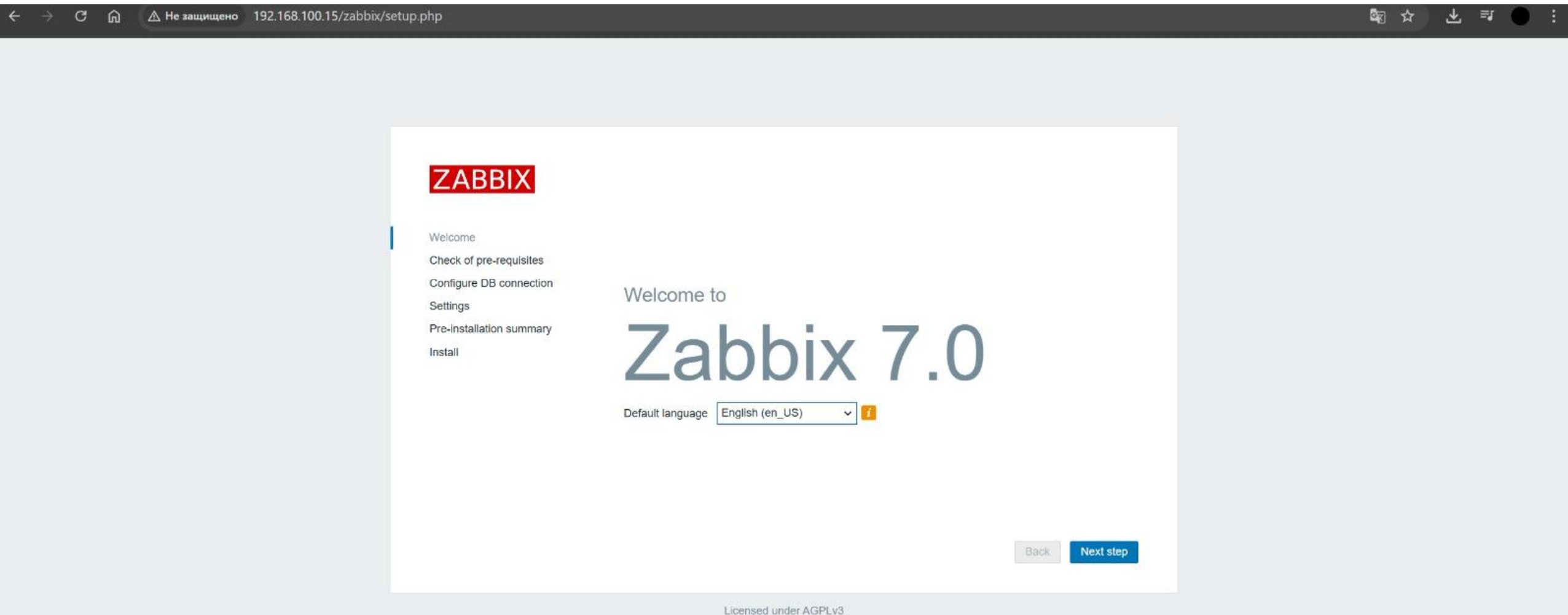
Showing 1 - 6 of 6 users

User details ↑	Email	Member since	Last login	Last activity	Last post	Days active	Messages posted	Actions
A Ali Melikov ali.m	ali.m@kei.local	2 days		2 days				Member ▾
I inal inal	hacizadeinal526@gmail.com	6 days		Today				System Admin ▾
I InalHacizade inalhacizade	inalhacizade@kei.local	6 days		2 days				Deactivated ▾
K kamran103 kamran103	agazadekamran12@gmail.com	6 days		6 days				System Admin ▾
R Ravan Hasanov ravan.h	ravan.h@kei.local	2 days		2 days				Member ▾
Y Yusif Axundiv yusif.a	yusif.a@kei.local	2 days		2 days				Member ▾

Showing 1 - 6 of 6 users

Show 10 rows per page

Zabbix (Monitoring)



Global view

All dashboards / Global view

Top hosts by CPU utilization

Host name	Utilization	1m avg	5m avg	15m avg	Processes
-----------	-------------	--------	--------	---------	-----------

[Zabbix server](#)

1.10↓
Zabbix server
Values per second

System information

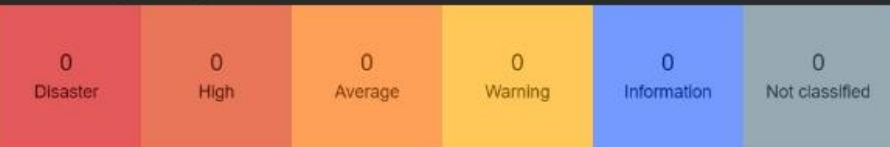
Parameter	Value	Details
Zabbix server is running	Yes	localhost:10051
Zabbix server version	7.0.23	
Zabbix frontend version	7.0.23	
Number of hosts (enabled/disabled)	1	1 / 0
Number of templates	354	
Number of items (enabled/disabled/not supported)	121	110 / 0 / 11
Number of triggers (enabled/disabled [problem/ok])	68	68 / 0 [0 / 68]

17:42
Baku

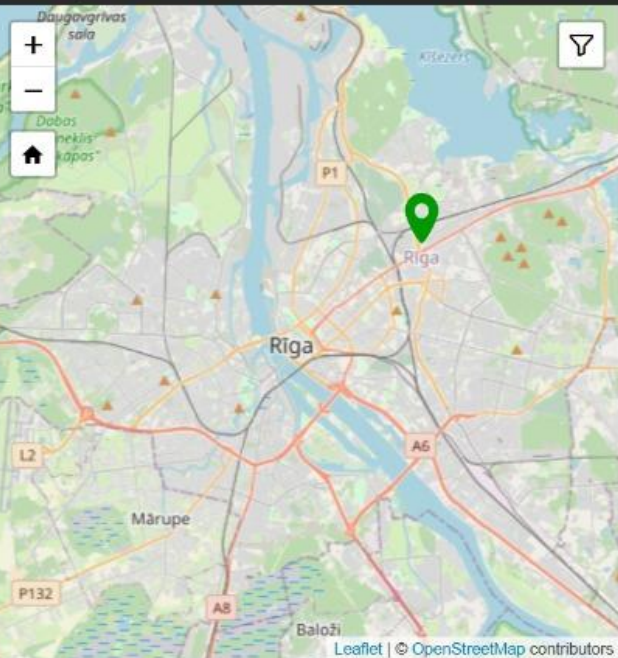
Host availability



Problems by severity



Geomap



Current problems

Time ▼	Info	Host	Problem • Severity	Duration	Update	Actions	Tags
No data found							

☐	Upsgenie	webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	OTRS CE	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	PagerDuty	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	Pushover	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	Redmine	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	Rocket.Chat	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	ServiceNow	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	SIGL4	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	Slack	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	SMS	SMS	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	SolarWinds Service Desk	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	SysAid	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	Telegram	Webhook	Enabled	5	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers, TelegramZabbix	Test
☐	TOPdesk	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	VictorOps	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	Zammad	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test
☐	Zendesk	Webhook	Disabled	4	Report not supported items, Report not supported low level discovery rules, Report problems to Zabbix administrators, Report unknown triggers	Test

GSM modem: "/dev/ttyS0"



Trigger actions ▾



Create action

Filter

Name

Status

Any

Enabled

Disabled

Apply

Reset

☐ Name ▲	Conditions	Operations	Status	Info
☐ Report problems to Zabbix administrators		Send message to user groups: Zabbix administrators via all media	Enabled	
☐ TelegramZabbix	Trigger severity is greater than or equals <i>Warning</i>	Send message to users: Admin (Zabbix Administrator) via Telegram Send message to user groups: Disabled, Enabled debug mode, Guests, Internal, No access to the frontend, Zabbix administrators via Telegram	Enabled	

Displaying 2 of 2 found

0 selected

Enable

Disable

Delete

Problem name: Linux: FS [/]: Space is low (used > 80%, total 19.5GB)
Host: MyServer
Severity: Warning
Operational data: Space used: 82.5%
Original problem ID: 199

5:51 PM

Problem: Linux: FS [/]: Space is low (used > 80%, total 19.5GB)
Problem started at 17:52:26 on 2026.02.14
Problem name: Linux: FS [/]: Space is low (used > 80%, total 19.5GB)
Host: nextcloud
Severity: Warning
Operational data: Space used: 82.5%
Original problem ID: 200

5:52 PM

Problem: Linux: FS [/]: Space is low (used > 80%, total 19.5GB)
Problem started at 17:52:26 on 2026.02.14
Problem name: Linux: FS [/]: Space is low (used > 80%, total 19.5GB)
Host: nextcloud
Severity: Warning
Operational data: Space used: 82.5%
Original problem ID: 200

5:52 PM

Zabbix_notif

Problem: NEW TEST TELEGRAM Problem started at 17:43:0...
Resolved in 18m 0s: NEW TEST TELEGRAM
Problem has been resolved in 18m 0s at 18:01:09 on 2026.02.14
Problem name: NEW TEST TELEGRAM
Host: MyServer
Severity: Not classified
Original problem ID: 197

6:01 PM

Problem: NEW TEST TELEGRAM
Problem started at 18:03:09 on 2026.02.14
Problem name: NEW TEST TELEGRAM
Host: MyServer
Severity: Not classified
Operational data: 0.042969
Original problem ID: 202

6:03 PM

 Write a message...

Zabbix_notif
bot



Message



Mute



Manage

@kel_zabbix_bot

Username



Add to Group



Bot Privacy Policy



Report



Stop and block bot

Nextcloud

Enter password:

Welcome to the MariaDB monitor. Commands end with ; or \g.

Your MariaDB connection id is 31

Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE nextcloud;

Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> CREATE USER 'nextcloud_user'@'localhost' IDENTIFIED BY '123';

Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON nextcloud.* TO 'nextcloud_user'@'localhost';

Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;

Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> EXIT;

Bye

kamran@kamran-VMware-Virtual-Platform:~\$ wget https://download.nextcloud.com/server/releases/latest.zip

--2026-02-12 03:19:13-- https://download.nextcloud.com/server/releases/latest.zip

Resolving download.nextcloud.com (download.nextcloud.com)... 5.9.202.145, 2a01:4f8:210:21c8::145

Connecting to download.nextcloud.com (download.nextcloud.com)|5.9.202.145|:443



Create administration account

Administration account name

Administration account password

▼ Storage & database

Data folder

/var/www/html/nextcloud/data

Database configuration

Only MySQL/MariaDB is available.
Install and activate additional PHP modules
to choose other database types.
For more details check out the
documentation. [🔗](#)

Database user

Database password

Database name

Database host

localhost

Please specify the port number along with the
host name (e.g., localhost:5432)



Рекомендованные приложения



Calendar

Составляйте графики работы и встреч, синхронизированные со всеми своими устройствами.



Contacts

Храните информацию о своих коллегах и друзьях в одном месте без утечки их личных данных.



Mail

Простое приложение электронной почты, прекрасно интегрированное с приложениями Файлы, Контакты и Календарь.



Nextcloud Office

Совместная работа с документами, таблицами и презентациями, основанная на Collabora Online.



Notes

Простое приложение для ведения заметок.



Talk

Общение в чате, видеозвонки, демонстрация экрана, онлайн-встречи и веб-конференции - в вашем браузере и на мобильных устройствах.



Пропустить

Установить рекомендуемые приложения

Zend Engine v4.3.6, Copyright (c) Zend Technologies

with Zend OPcache v8.3.6, Copyright (c), by Zend Technologies

kamran@kamran-VMware-Virtual-Platform:~\$ sudo nano /etc/php/8.1/apache2/php.ini

kamran@kamran-VMware-Virtual-Platform:~\$ sudo nano /etc/p

pam.conf pcmcia/ pm/ profile python3.12/

pam.d/ perl/ pnm2ppa.conf profile.d/

papersize php/ polkit-1/ protocols

passwd pki/ ppp/ pulse/

passwd- plymouth/ printcap python3/

kamran@kamran-VMware-Virtual-Platform:~\$ sudo nano /etc/php/8.3/apache2/php.ini

kamran@kamran-VMware-Virtual-Platform:~\$ sudo systemctl restart apache2

kamran@kamran-VMware-Virtual-Platform:~\$ sudo nano /etc/apache2/sites-enabled/di

sk.rio-zbx.ru.conf

kamran@kamran-VMware-Virtual-Platform:~\$ sudo systemctl restart apache2

kamran@kamran-VMware-Virtual-Platform:~\$ sudo nano /etc/apache2/sites-enabled/disk.rio-zbx.ru

kamran@kamran-VMware-Virtual-Platform:~\$ sudo systemctl restart apache2

kamran@kamran-VMware-Virtual-Platform:~\$ sudo -u www-data php /var/www/html/nextcloud/occ cor

System config value trusted_domains => 2 set to string nextcloud.kei.local

kamran@kamran-VMware-Virtual-Platform:~\$ sudo -u www-data php /var/www/html/nextcloud/occ cor

System config value trusted_domains => 3 set to string kei.local

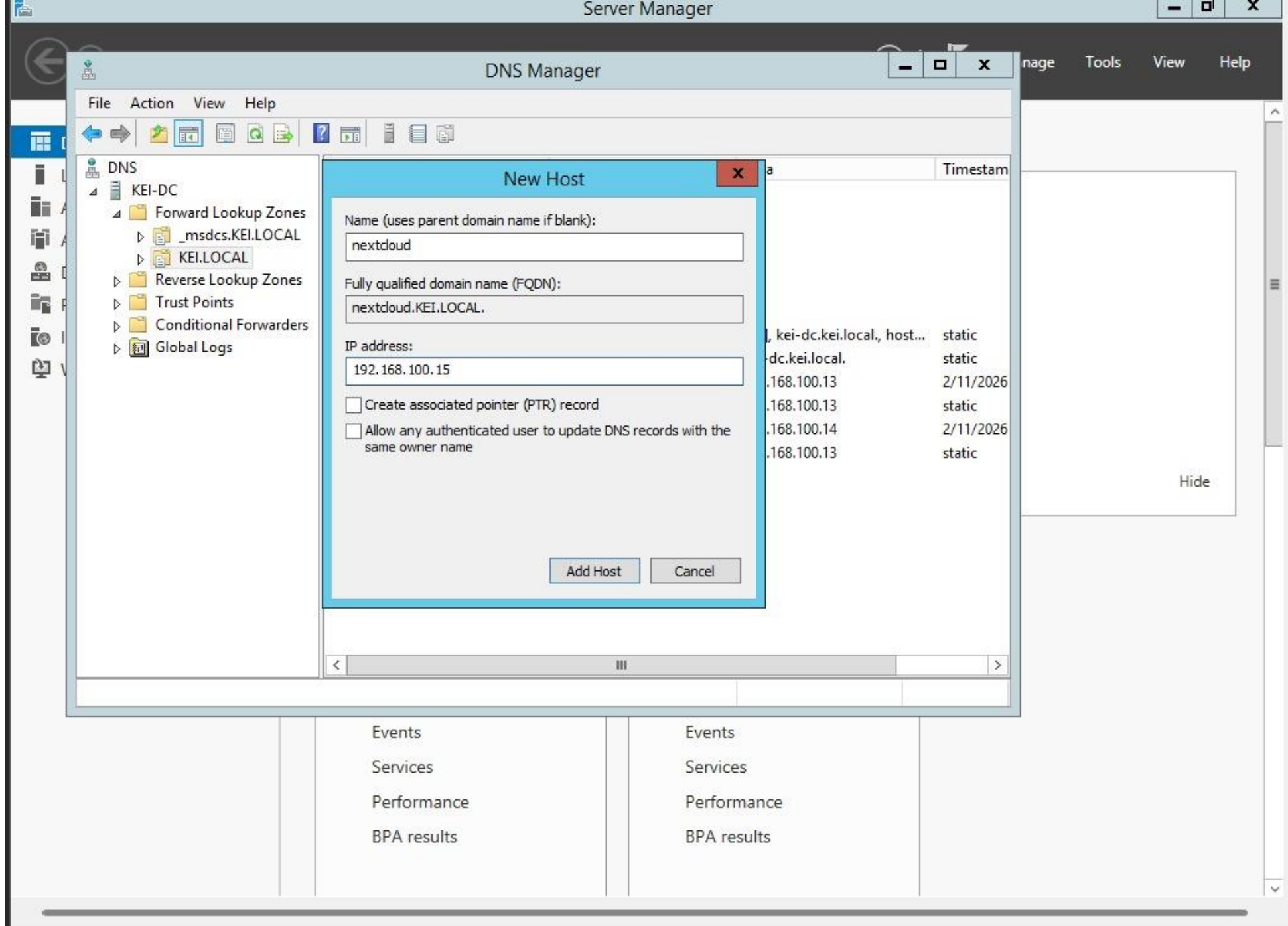
kamran@kamran-VMware-Virtual-Platform:~\$ sudo -u www-data php /var/www/html/nextcloud/occ cor

192.168.100.15

nextcloud.kei.local

kei.local

kamran@kamran-VMware-Virtual-Platform:~\$



kei.local

kamran@kamran-VMware-Virtual-Platform:~\$ nslookup nextcloud.kei.local

Server: 192.168.100.13

Address: 192.168.100.13#53

Name: nextcloud.kei.local

Address: 192.168.100.15

kamran@kamran-VMware-Virtual-Platform:~\$

Добрый день

📁 Рекомендуемые фа...

- 

Birdie.jpg
Последнее обновление: 12 ча...
- 

Steps.jpg
Последнее обновление: 12 ча...
- 

Library.jpg
Последнее обновление: 12 ча...
- 

Vineyard.jpg
Последнее обновление: 12 ча...
- 

Readme.md
Последнее обновление: 12 ча...
- 

Toucan.jpg
Последнее обновление: 12 ча...
- 

Frog.jpg
Последнее обновление: 12 ча...

🔍 Упоминания в обсу...

- 

Нет ни одного
непрочитанного упоминания



Давайте начнём!
⚙ Управление настройкам...



Обновления приложения «Ко...
Новое в приложении «Конф...



Личные заметки
Обсуждение создано системой...
- ## ✉ Важная почта
- 

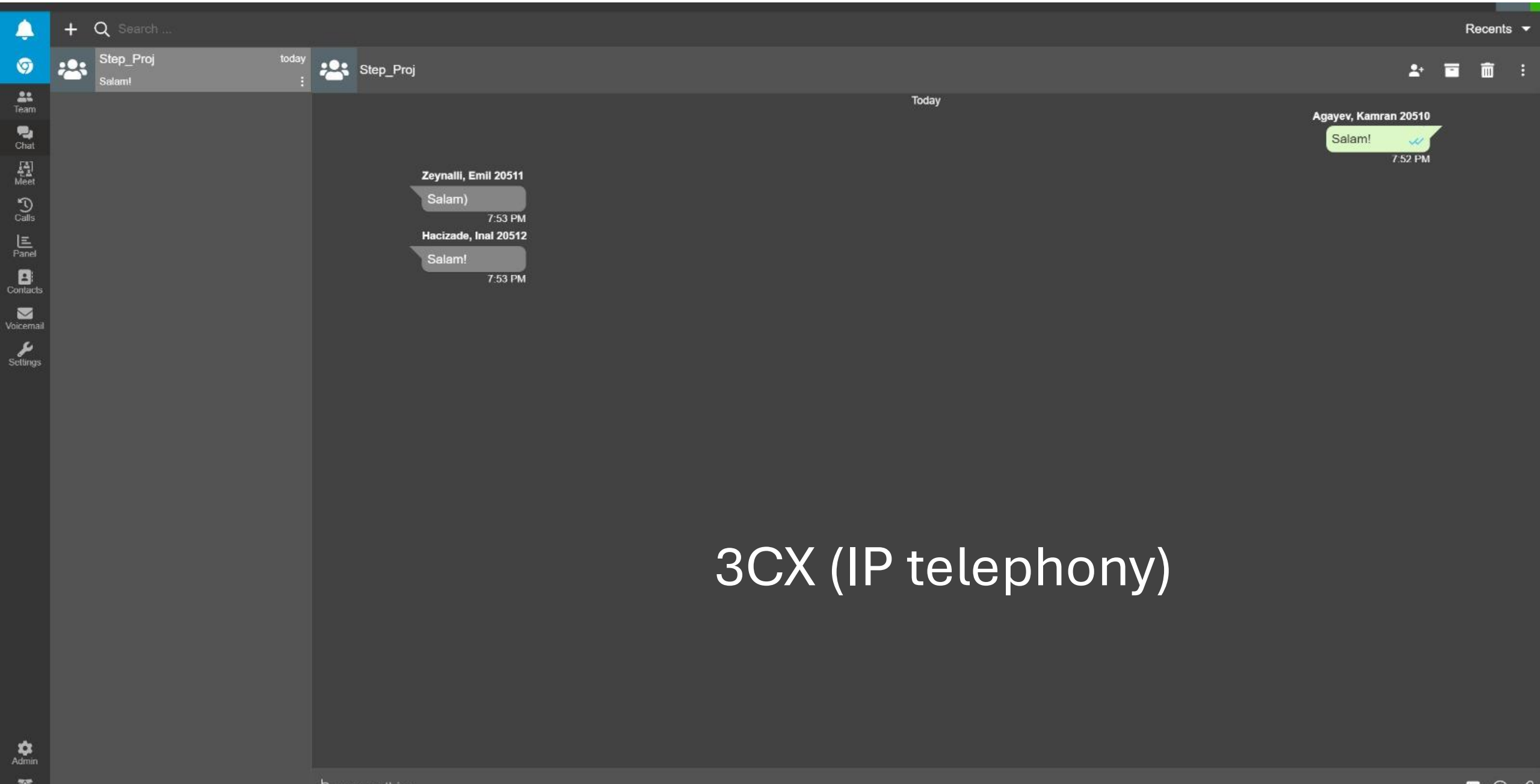
Сообщений ещё нет

Настроить учётную запись
- ## 📅 Приближающиеся ...
- 

На сегодня больше нет событий



Пример события — отroyте м...
через 6 дней
- ✎ Настроить



3CX (IP telephony)



  Search ...

All ▾



HI	Hacizade, Inal 20512 02/09/2026 7:55 PM, 00:01:30	  
----	--	---



HI	Hacizade, Inal 20512 02/09/2026 7:55 PM, 00:00:14	  
----	--	---



HI	Hacizade, Inal 20512 02/09/2026 7:54 PM, 00:00:22	  
----	--	---





20510
Agayev, Kamran

00:07 - Соединено



Новый вызов



Конференция



Перевод



Наушник



Набор



Видео



Запись



Удержать



Откл. звук



```
#!/bin/bash
```

```
SWITCH_IP="10.1.99.254"
```

```
USER="admin"
```

```
PASS="password"
```

```
BACKUP_DIR="/opt/switch_backups"
```

```
DATE=$(date +%F_%H-%M-%S)
```

```
mkdir -p $BACKUP_DIR
```

```
/usr/bin/expect <<EOF > $BACKUP_DIR/core-switch_$DATE.cfg
```

```
set timeout 20
```

```
spawn ssh -o StrictHostKeyChecking=no $USER@$SWITCH_IP
```

```
expect "password:"
```

```
send "$PASS\r"
```

```
expect ">"
```

```
send "enable\r"
```

```
expect "Password:"
```

```
send "$PASS\r"
```

```
expect "#"
```

```
send "terminal length 0\r"
```

```
expect "#"
```

```
send "show running-config\r"
```

```
expect "#"
```

Backing up switch configurations

```
root@kali:~#
root@kali:~#
root@kali:~#
root@kali:~#
root@kali:~#
root@kali:~#
root@kali:~#
root@kali:~#
root@kali:~#
root@kali:~# nano /opt/backup_switch.sh
root@kali:~# apt install expect -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
expect is already the newest version (5.45.4-2+b1).
expect set to manually installed.
The following packages were automatically installed and are no longer required:
 automater bdfproxy blindelephant creddump deblaze dnmap fimap findmyhash funkload
 ghost-phisher girl1.2-clutter-gst-3.0 girl1.2-gtkclutter-1.0 girl1.2-mutter-3
 gnome-theme-kali grabber gtk2-engines-murrine hexorbase intersect keimpx leafpad
 libadns1 libbrlapi0.6 libgeos-3.7.2 libgnome-desktop-3-17 libgssglue1 libhavege1
 libigdgmm9 libilmbase23 libisl19 libjaxp1.3-java liblinear3 liblirc-client0
 libllvm8 liblouis17 libmutter-3-0 libopenexr23 libproj13 libqscintilla2-qt4-13
 libqscintilla2-qt4-110n libradare2-3.2 libsnmp30 libvpx5 libwxgtk3.0-0v5
 libx265-165 libx265-176 libxdot4 maltego-teeth miranda nfs Spy pdgmail powerfuzzer
 proxystrike python-adns python-configargparse python-configobj python-construct
 python-dns python-dnslib python-easygui python-fuse python-gtkspellcheck
 python-h2 python-hpack python-httplib2 python-hyperframe python-jsonpickle
 python-ldap3 python-ldapdomaindump python-levenshtein python-m2crypto
 python-mechanize python-metaconfig python-msgpack python-pcap python-pyperclip
 python-pyqt5 python-qscintilla2 python-slugify python-twisted-web python-urwid
 python-webunit python-wxtools ruby-librex sslcaudit sslstrip tcpwatch-http-proxy
 tpm2-abrmd tpm2-tools u3-pwn u3-tool ua-tester wifitap wol-e
Use 'apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 39 not upgraded.
root@kali:~#
```

```
root@kali:/opt/switch_backups# crontab -e
no crontab for root - using an empty one
```

Select an editor. To change later, run 'select-editor'.

1. /usr/bin/vim.gtk3
2. /bin/nano <----- easiest
3. /usr/bin/vim.basic
4. /usr/bin/mcedit
5. /usr/bin/vim.tiny

Choose 1-5 [2]: 2

crontab: installing new crontab

```
root@kali:/opt/switch_backups#
```

```
root@kali:/opt/switch_backups# crontab -l
```

```
# Edit this file to introduce tasks to be run by cron.
```

```
#
```

```
# Each task to run has to be defined through a single line
```

```
# indicating with different fields when the task will be run
```

```
# and what command to run for the task
```

```
* * * * * /opt/backup_switch.sh
```

```
# To define the time you can provide concrete values for
```

```
# minute (m), hour (h), day of month (dom), month (mon),
```

```
# and day of week (dow) or use '*' in these fields (for 'any').
```

```
#
```

```
# Notice that tasks will be started based on the cron's system
```

```
# daemon's notion of time and timezones.
```

```
#
```

```
# Output of the crontab jobs (including errors) is sent through
```

```
# email to the user the crontab file belongs to (unless redirected).
```

```
#
```

```
# For example, you can run a backup of all your user accounts
```

```
# at 5 a.m every week with:
```

```
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
```

```
#
```

```
# For more information see the manual pages of crontab(5) and cron(8)
```



```
# Edit this file to introduce tasks to be run by cron.
```

```
#
```

```
# Each task to run has to be defined through a single line
```

```
# indicating with different fields when the task will be run
```

```
# and what command to run for the task
```

```
* * * * /opt/backup_switch.sh
```

```
# To define the time you can provide concrete values for
```

```
# minute (m), hour (h), day of month (dom), month (mon),
```

```
# and day of week (dow) or use '*' in these fields (for 'any').
```

```
#
```

```
# Notice that tasks will be started based on the cron's system
```

```
# daemon's notion of time and timezones.
```

```
#
```

```
# Output of the crontab jobs (including errors) is sent through
```

```
# email to the user the crontab file belongs to (unless redirected).
```

```
#
```

```
# For example, you can run a backup of all your user accounts
```

```
# at 5 a.m every week with:
```

```
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
```

```
#
```

```
# For more information see the manual pages of crontab(5) and cron(8)
```

```
#
```

```
# m h dom mon dow  command
```

```
^G Get Help
^X Exit
```

```
^O Write Out
^R Read File
```

```
^W Where Is
^_ Replace
```

```
^K Cut Text
^U Paste Text
```

```
^J Justify
^T To Spell
```

```
^C Cur Pos
^_ Go To Line
```

```
M-U Undo
M-E Redo
```

```
M-A Mark Text
M-6 Copy Text
```

```
M-] To Bracket
^C Where Was
```

```
M-; Previous
M-~ Next
```



```
root@kali:/opt/switch_backups# ls /opt/switch_backups/
core-switch_2026-02-17_10-51-01.cfg  core-switch_2026-02-17_10-54-01.cfg
root@kali:/opt/switch_backups# systemctl status cron
● cron.service - Regular background program processing daemon
   Loaded: loaded (/lib/systemd/system/cron.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-02-17 10:40:18 EST; 14min ago
     Docs: man:cron(8)
 Main PID: 415 (cron)
    Tasks: 7 (limit: 4648)
   Memory: 3.9M
    CGroup: /system.slice/cron.service
            └─ 415 /usr/sbin/cron -f
               └─ 2187 /usr/sbin/CRON -f
                  └─ 2188 /bin/sh -c /opt/backup_switch.sh
                     └─ 2189 /bin/bash /opt/backup_switch.sh
                        └─ 2192 /usr/bin/expect
                           └─ 2193 ssh -o StrictHostKeyChecking=no admin@10.1.99.254
```

```
Feb 17 10:40:18 kali systemd[1]: Started Regular background program processing daemon.
Feb 17 10:40:18 kali cron[415]: (CRON) INFO (pidfile fd = 3)
Feb 17 10:40:18 kali cron[415]: (CRON) INFO (Running @reboot jobs)
Feb 17 10:45:01 kali CRON[1496]: pam_unix(cron:session): session opened for user root
Feb 17 10:45:01 kali CRON[1497]: (root) CMD (command -v debian-sa1 > /dev/null && de
Feb 17 10:45:01 kali CRON[1496]: pam_unix(cron:session): session closed for user root
Feb 17 10:54:01 kali CRON[2187]: pam_unix(cron:session): session opened for user root
Feb 17 10:54:01 kali CRON[2188]: (root) CMD (/opt/backup_switch.sh)
```

```
root@kali:/opt/switch_backups# ^C
root@kali:/opt/switch_backups# ls /opt/switch_backups/
core-switch_2026-02-17_10-51-01.cfg  core-switch_2026-02-17_10-54-01.cfg
root@kali:/opt/switch_backups# ls /opt/switch_backups/
core-switch_2026-02-17_10-51-01.cfg  core-switch_2026-02-17_10-55-01.cfg
core-switch_2026-02-17_10-54-01.cfg
root@kali:/opt/switch_backups# ls /opt/switch_backups/
core-switch_2026-02-17_10-51-01.cfg  core-switch_2026-02-17_10-55-01.cfg
core-switch_2026-02-17_10-54-01.cfg
root@kali:/opt/switch_backups# ls /opt/switch_backups/
```

```
root@kali:/opt/switch_backups# ls -lh
```

```
total 28K
```

```
-rw-r--r-- 1 root root 67 Feb 17 10:51 core-switch_2026-02-17_10-51-01.cfg  
-rw-r--r-- 1 root root 196 Feb 17 10:56 core-switch_2026-02-17_10-54-01.cfg  
-rw-r--r-- 1 root root 196 Feb 17 10:57 core-switch_2026-02-17_10-55-01.cfg  
-rw-r--r-- 1 root root 196 Feb 17 10:58 core-switch_2026-02-17_10-56-01.cfg  
-rw-r--r-- 1 root root 131 Feb 17 10:59 core-switch_2026-02-17_10-57-01.cfg  
-rw-r--r-- 1 root root 85 Feb 17 10:59 core-switch_2026-02-17_10-58-01.cfg  
-rw-r--r-- 1 root root 57 Feb 17 10:59 core-switch_2026-02-17_10-59-01.cfg
```

```
root@kali:/opt/switch_backups#
```